INCH-POUND

MIL-M-28787/435
29 March 1991

# MILITARY SPECIFICATION

## MODULES, STANDARD ELECTRONIC

## MEMORY ADDRESS MODULE, KEY CODE JMM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

### 1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic memory address module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).



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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.944 failures per million operating hours for class I or 4.859 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be JMM.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.



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TABLE I. Module electrical requirements.

| Test cond | Parameter 5/ | Ref test table | Test type 1/-3/ | Class I and II 25 ± 5°C 4/ | | Class I 0 ± 5°C 4/ 60 ± 5°C 4/ | | Class II -55 ± 5°C 4/ 85 ± 5°C 4/ | | Units |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Min | Max | Min | Max | Min | Max | |
| A | Isolation | II | S | 10 | INF | 10 | INF | 10 | INF | MOhm |
| B1-B6 | Continuity | III | X | 0 | 1 | 0 | 1 | 0 | 1 | Ohm |
| C | Vcc stress | IV | 0 | ----------Module shall pass all remaining tests---------- | | | | | | |
| D1 | Ibd | V | S | 0.0 | 0.8 | 0.0 | 0.8 | 0.0 | 0.8 | mA |
| D2 | Ibd | V | S | 0.0 | 0.3 | 0.0 | 0.3 | 0.0 | 0.3 | mA |
| D3-D14 | Ibd | V | S | 0.0 | 0.2 | 0.0 | 0.2 | 0.0 | 0.2 | mA |
| D15 | Ibd | V | S | 0.0 | 0.1 | 0.0 | 0.1 | 0.0 | 0.1 | mA |
| D16 | Ibd | V | S | 0.0 | 0.2 | 0.0 | 0.2 | 0.0 | 0.2 | mA |
| D17 | Ibd | V | S | 0 0 | 0.7 | 0.0 | 0.7 | 0.0 | 0 7 | mA |
| D18 | Ibd | V | S | 0 0 | 0.5 | 0.0 | 0.5 | 0.0 | 0 5 | mA |
| D19-D26 | Ibd | V | S | 0.0 | 0.2 | 0.0 | 0.2 | 0.0 | 0.2 | mA |
| D27, D28 | Ibd | V | S | 0.0 | 0.1 | 0.0 | 0.1 | 0.0 | 0.1 | mA |
| D29 | Ibd | V | S | -0.057 | 0.16 | -0.057 | 0.16 | -0.057 | 0.16 | mA |
| D30 | Ibd | V | S | 0 0 | 0.1 | 0.0 | 0.1 | 0.0 | 0 1 | mA |
| D31 | Ibd | V | S | 0 0 | 0.3 | 0.0 | 0.3 | 0.0 | 0 3 | mA |
| E1 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E2, E3 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E4 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E5, E6 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E7 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100 0 | mA |
| E8, E9 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E10 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E11-E16 | Ios | VI | S | -150 0 | -60.0 | -150.0 | -60.0 | -150.0 | -60 0 | mA |
| E17 | Ios | VI | S | -325 0 | -100.0 | -325.0 | -100.0 | -325.0 | -100 0 | mA |
| E18, E19 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E20 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E21, E22 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E23 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E24, E25 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E26 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |

- See footnotes at end of table.



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TABLE I. Module electrical requirements - Continued

| Test cond | Parameter 5/ | Ref test table | Test type 1/-3/ | Class I and II 25 ± 5°C 4/ | | Class I 0 ± 5°C 4/ 60 ± 5°C 4/ | | Class II -55 ± 5°C 4/ 85 ± 5°C 4/ | | Units |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Min | Max | Min | Max | Min | Max | |
| E27-E29 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E30,E31 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E32 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E33,E34 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E35-E39 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E40,E41 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E42 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E43-E45 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E46,E47 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E48 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E49,E50 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E51 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E52,E53 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E54 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E55-E57 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E58 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E59,E60 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E61 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E62,E63 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E64 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E65,E66 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E67 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E68,E69 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E70 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E71 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E72,E73 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E74 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E75,E76 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E77 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E78,E79 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E80 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E81 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E82-E85 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E86 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |

- See footnotes at end of table





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TABLE I Module electrical requirements - Continued

| Test cond | Parameter 5/ | Ref test table | Test type 1/-3/ | Class I and II 25 ± 5°C 4/ | | Class I 0 ± 5°C 4/ 60 ± 5°C 4/ | | Class II -55 ± 5°C 4/ 85 ± 5°C 4/ | | Units |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Min | Max | Min | Max | Min | Max | |
| E87 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E88,E89 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E90 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E91,E92 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E93 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E94 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E95 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E96-E99 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E100 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E101,E102 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E103-E108 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E109,E110 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E111 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E112,E113 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E114 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E115 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E116 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E117,E118 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E119 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E120,E121 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E122 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E123,E124 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E125 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E126 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E127 | Ios | VI | S | -100.0 | -20.0 | -100.0 | -20.0 | -100.0 | -20.0 | mA |
| E128 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E129 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E130,E131 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E132 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E133,E134 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E135 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E136,E137 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |
| E138 | Ios | VI | S | -150.0 | -60.0 | -150.0 | -60.0 | -150.0 | -60.0 | mA |
| E139 | Ios | VI | S | -325.0 | -100.0 | -325.0 | -100.0 | -325.0 | -100.0 | mA |

- See footnotes at end of table.



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TABLE I. Module electrical requirements - Continued

| Test cond | Parameter 5/ | Ref test table | Test type 1/-3/ | Class I and II 25 ± 5°C 4/ | | Class I 0 ± 5°C 4/ 60 ± 5°C 4/ | | Class II -55 ± 5°C 4/ 85 ± 5°C 4/ | | Units |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Min | Max | Min | Max | Min | Max | |
| F1-F8 | Ith | VII | X | -1.35 | 0.19 | -1.35 | 0.19 | -1.35 | 0.19 | mA |
| F9,F10 | Ith | VII | X | -0.75 | 0.17 | -0.75 | 0.17 | -0.75 | 0.17 | mA |
| F11 | Ith | VII | X | -4.8 | 0.18 | -4.8 | 0.18 | -4.8 | 0.16 | mA |
| F12 | Ith | VII | X | -1.8 | 0.06 | -1.8 | 0.06 | -1.8 | 0.06 | mA |
| F13-F24 | Ith | VII | X | -1.2 | 0.04 | -1.2 | 0.04 | -1.2 | 0.04 | mA |
| F25,F26 | Ith | VII | X | -0.75 | 0.17 | -0.75 | 0.17 | -0.75 | 0.17 | mA |
| F27-F30 | Ith | VII | X | -1.35 | 0.19 | -1.35 | 0.19 | -1.35 | 0.19 | mA |
| F31 | Ith | VII | X | -0.6 | 0.02 | -0.6 | 0.02 | -0.6 | 0.02 | mA |
| F32 | Ith | VII | X | -1.2 | 0.04 | -1.2 | 0.04 | -1.2 | 0.04 | mA |
| F33-F36 | Ith | VII | X | -1.35 | 0.19 | -1.35 | 0.19 | -1.35 | 0.19 | mA |
| F37-F46 | Ith | VII | X | -0.75 | 0.17 | -0.75 | 0.17 | -0.75 | 0.17 | mA |
| F47 | Ith | VII | X | -4.2 | 0.14 | -4.2 | 0.14 | -4.2 | 0.14 | mA |
| F48 | Ith | VII | X | -3.0 | 0.1 | -3.0 | 0.1 | -3.0 | 0.1 | mA |
| F49-F56 | Ith | VII | X | -1.2 | 0.04 | -1.2 | 0.04 | -1.2 | 0.04 | mA |
| F57,F58 | Ith | VII | X | -0.6 | 0.02 | -0.6 | 0.02 | -0.6 | 0.02 | mA |
| F59 | Ith | VII | X | -3.31 | -1.41 | -3.31 | -1.41 | -3.31 | -1.41 | mA |
| F60-F65 | Ith | VII | X | -0.75 | 0.17 | -0.75 | 0.17 | -0.75 | 0.17 | mA |
| F66 | Ith | VII | X | -0.6 | 0.02 | -0.6 | 0.02 | -0.6 | 0.02 | mA |
| F67 | Ith | VII | X | -1.8 | 0.06 | -1.8 | 0.06 | -1.8 | 0.06 | mA |
| F68-F71 | Ith | VII | X | -0.75 | 0.17 | -0.75 | 0.17 | -0.75 | 0.17 | mA |
| G1-G8 | Iil | VIII | X | -1.35 | 0.19 | -1.35 | 0.19 | -1.35 | 0.19 | mA |
| G9,G10 | Iil | VIII | X | -0.75 | 0.17 | -0.75 | 0.17 | -0.75 | 0.17 | mA |
| G11 | Iil | VIII | X | -4.8 | 0.16 | -4.8 | 0.16 | -4.8 | 0.16 | mA |
| G12 | Iil | VIII | X | -1.8 | 0.06 | -1.8 | 0.06 | -1.8 | 0.06 | mA |
| G13-G24 | Iil | VIII | X | -1.2 | 0.04 | -1.2 | 0.04 | -1.2 | 0.04 | mA |
| G25,G26 | Iil | VIII | X | -0.75 | 0.17 | -0.75 | 0.17 | -0.75 | 0.17 | mA |
| G27-G30 | Iil | VIII | X | -1.35 | 0.19 | -1.35 | 0.19 | -1.35 | 0.19 | mA |
| G31 | Iil | VIII | X | -0.6 | 0.02 | -0.6 | 0.02 | -0.6 | 0.02 | mA |
| G32 | Iil | VIII | X | -1.2 | 0.04 | -1.2 | 0.04 | -1.2 | 0.04 | mA |
| G33-G36 | Iil | VIII | X | -1.35 | 0.19 | -1.35 | 0.19 | -1.35 | 0.19 | mA |
| G37-G46 | Iil | VIII | X | -0.75 | 0.17 | -0.75 | 0.17 | -0.75 | 0.17 | mA |
| G47 | Iil | VIII | X | -4.2 | 0.14 | -4.2 | 0.14 | -4.2 | 0.14 | mA |
| G48 | Iil | VIII | X | -3.0 | 0.1 | -3.0 | 0.1 | -3.0 | 0.1 | mA |

- See footnotes at end of table.



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TABLE I. Module electrical requirements - Continued.

| Test cond | Parameter 5/ | Ref test table | Test type 1/-3/ | Class I and II 25 ± 5°C 4/ Min | Class I and II 25 ± 5°C 4/ Max | Class I 0 ± 5°C 4/ 60 ± 5°C 4/ Min | Class I 0 ± 5°C 4/ 60 ± 5°C 4/ Max | Class II -55 ± 5°C 4/ 85 ± 5°C 4/ Min | Class II -55 ± 5°C 4/ 85 ± 5°C 4/ Max | Units |
|---|---|---|---|---|---|---|---|---|---|---|
| G49-G56 | Iil | VIII | X | -1.2 | 0.04 | -1.2 | 0.04 | -1.2 | 0.04 | mA |
| G57,G58 | Iil | VIII | X | -0.6 | 0.02 | -0.6 | 0.02 | -0.6 | 0.02 | mA |
| G59 | Iil | VIII | X | -3.31 | -1.41 | -3.31 | -1.41 | -3.31 | -1.41 | mA |
| G60-G65 | Iil | VIII | X | -0.75 | 0.17 | -0.75 | 0.17 | -0.75 | 0.17 | mA |
| G66 | Iil | VIII | X | -0.6 | 0.02 | -0.6 | 0.02 | -0.6 | 0.02 | mA |
| G67 | Iil | VIII | X | -1.8 | 0.06 | -1.8 | 0.06 | -1.8 | 0.06 | mA |
| G68-G71 | Iil | VIII | X | -0.75 | 0.17 | -0.75 | 0.17 | -0.75 | 0.17 | mA |
| H1-H139 | Voh | IX | X | 2.4 | 4.5 | 2.4 | 4.5 | 2.4 | 4.5 | V |
| J1-J139 | Vol | X | X | 0.0 | 0.5 | 0.0 | 0.5 | 0.0 | 0.5 | V |
| K1-K16 | Iozh | XI | X | -0.05 | 0.05 | -0.05 | 0.05 | -0.05 | 0.05 | mA |
| K17-K32 | Iozl | XI | X | -0.05 | 0.05 | -0.05 | 0.05 | -0.05 | 0.05 | mA |
| L1-L31 | Vic | XII | S | -1.2 | 0.0 | -1.2 | 0.0 | -1.2 | 0.0 | V |
| M1 | PSFC | XIII | S | 5.6 | 10.4 | 5.6 | 10.4 | 5.6 | 10.4 | uF |
| M2 | PSFC | XIII | S | 7.0 | 13.0 | 7.0 | 13.0 | 7.0 | 13.0 | uF |
| N | Icc | XIV | X | 680 | 1633 | 680 | 1633 | 680 | 1633 | mA |
| P1,P2 | Functional | XV | X | See table XV | | | | | | |
| Q1,Q2 | Dynamic | XVI | S | See table XVI | | | | | | |
| Q3,Q4 | Dynamic | XVI | Q | See table XVI | | | | | | |
| R1 | Ci/o | XVIII | Q | 12 | 216 | 12 | 216 | 12 | 216 | pF |
| R2 | Ci/o | XVIII | Q | 6 | 100 | 6 | 100 | 6 | 100 | pF |
| R3-R10 | Ci/o | XVIII | Q | 4 | 100 | 4 | 100 | 4 | 100 | pF |
| R11-R14 | Ci/o | XVIII | Q | 4 | 75 | 4 | 75 | 4 | 75 | pF |
| R15 | Ci/o | XVIII | Q | 2 | 75 | 2 | 75 | 2 | 75 | pF |
| R16 | Ci/o | XVIII | Q | 2 | 100 | 2 | 100 | 2 | 100 | pF |
| R17 | Ci/o | XVIII | Q | 10 | 200 | 10 | 200 | 10 | 200 | pF |

- See footnotes at end of table.



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TABLE I. Module electrical requirements - Continued.

| Test cond | Parameter 5/ | Ref test table | Test type 1/-3/ | Class I and II 25 ± 5°C 4/ | | Class I 0 ± 5°C 4/ 60 ± 5°C 4/ | | Class II -55 ± 5°C 4/ 85 ± 5°C 4/ | | Units |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Min | Max | Min | Max | Min | Max | |
| R18 | C1/D | XVIII | Q | 10 | 125 | 10 | 125 | 10 | 125 | pF |
| R19-R26 | C1/D | XVIII | Q | 4 | 100 | 4 | 100 | 4 | 100 | pF |
| R27,R28 | C1/D | XVIII | Q | 2 | 75 | 2 | 75 | 2 | 75 | pF |
| R29 | C1/D | XVIII | Q | 4 | 75 | 4 | 75 | 4 | 75 | pF |
| R30 | C1/D | XVIII | Q | 2 | 75 | 2 | 75 | 2 | 75 | pF |
| R31 | C1/D | XVIII | Q | 6 | 125 | 6 | 125 | 6 | 125 | pF |
| R32-R35 | C1/D | XVIII | Q | 10 | 125 | 10 | 125 | 10 | 125 | pF |
| R36-R39 | C1/D | XVIII | Q | 2 | 75 | 2 | 75 | 2 | 75 | pF |
| R40-R43 | C1/D | XVIII | Q | 10 | 125 | 10 | 125 | 10 | 125 | pF |
| R44,R45 | C1/D | XVIII | Q | 6 | 125 | 6 | 125 | 6 | 125 | pF |
| R46-R49 | C1/D | XVIII | Q | 2 | 75 | 2 | 75 | 2 | 75 | pF |
| R50 | C1/D | XVIII | Q | 6 | 125 | 6 | 125 | 6 | 125 | pF |
| R51 | C1/D | XVIII | Q | 6 | 150 | 6 | 150 | 6 | 150 | pF |
| R52 | C1/D | XVIII | Q | 10 | 150 | 10 | 150 | 10 | 150 | pF |
| R53 | C1/D | XVIII | Q | 10 | 125 | 10 | 125 | 10 | 125 | pF |
| R54 | C1/D | XVIII | Q | 10 | 100 | 10 | 100 | 10 | 100 | pF |
| R55-R59 | C1/D | XVIII | Q | 10 | 125 | 10 | 125 | 10 | 125 | pF |
| R60-R63 | C1/D | XVIII | Q | 8 | 100 | 8 | 100 | 8 | 100 | pF |
| R64-R67 | C1/D | XVIII | Q | 2 | 75 | 2 | 75 | 2 | 75 | pF |
| R68 | C1/D | XVIII | Q | 8 | 125 | 8 | 125 | 8 | 125 | pF |
| R69-R71 | C1/D | XVIII | Q | 8 | 100 | 8 | 100 | 8 | 100 | pF |
| R72 | C1/D | XVIII | Q | 4 | 100 | 4 | 100 | 4 | 100 | pF |
| R73 | C1/D | XVIII | Q | 4 | 150 | 4 | 150 | 4 | 150 | pF |
| R74-R77 | C1/D | XVIII | Q | 2 | 75 | 2 | 75 | 2 | 75 | pF |
| R78 | C1/D | XVIII | Q | 4 | 214 | 4 | 214 | 4 | 214 | pF |
| R79 | C1/D | XVIII | Q | 4 | 75 | 4 | 75 | 4 | 75 | pF |
| R80 | C1/D | XVIII | Q | 8 | 100 | 8 | 100 | 8 | 100 | pF |
| R81 | C1/D | XVIII | Q | 8 | 125 | 8 | 125 | 8 | 125 | pF |
| R82-R85 | C1/D | XVIII | Q | 8 | 100 | 8 | 100 | 8 | 100 | pF |
| R86,R87 | C1/D | XVIII | Q | 8 | 125 | 8 | 125 | 8 | 125 | pF |

- See footnotes at end of table.



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TABLE I. Module electrical requirements - Continued.

| Test cond | Parameter 5/ | Ref test table | Test type 1/-3/ | Class I and II 25 ± 5°C 4/ | | Class I 0 ± 5°C 4/ 60 ± 5°C 4/ | | Class II -55 ± 5°C 4/ 85 ± 5°C 4/ | | Units |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Min | Max | Min | Max | Min | Max | |
| S | Input clamp diode latch-up | XIX | Q | -1.2 | 2.4 | -1.2 | 2.4 | -1.2 | 2.4 | V |
| T1,T2 | PSTR | XX | Q | 0.0 | 400 | 0.0 | 400 | 0.0 | 400 | mV |

- See footnotes on next page.



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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.



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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/435-1 for class I and M28787/435-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.



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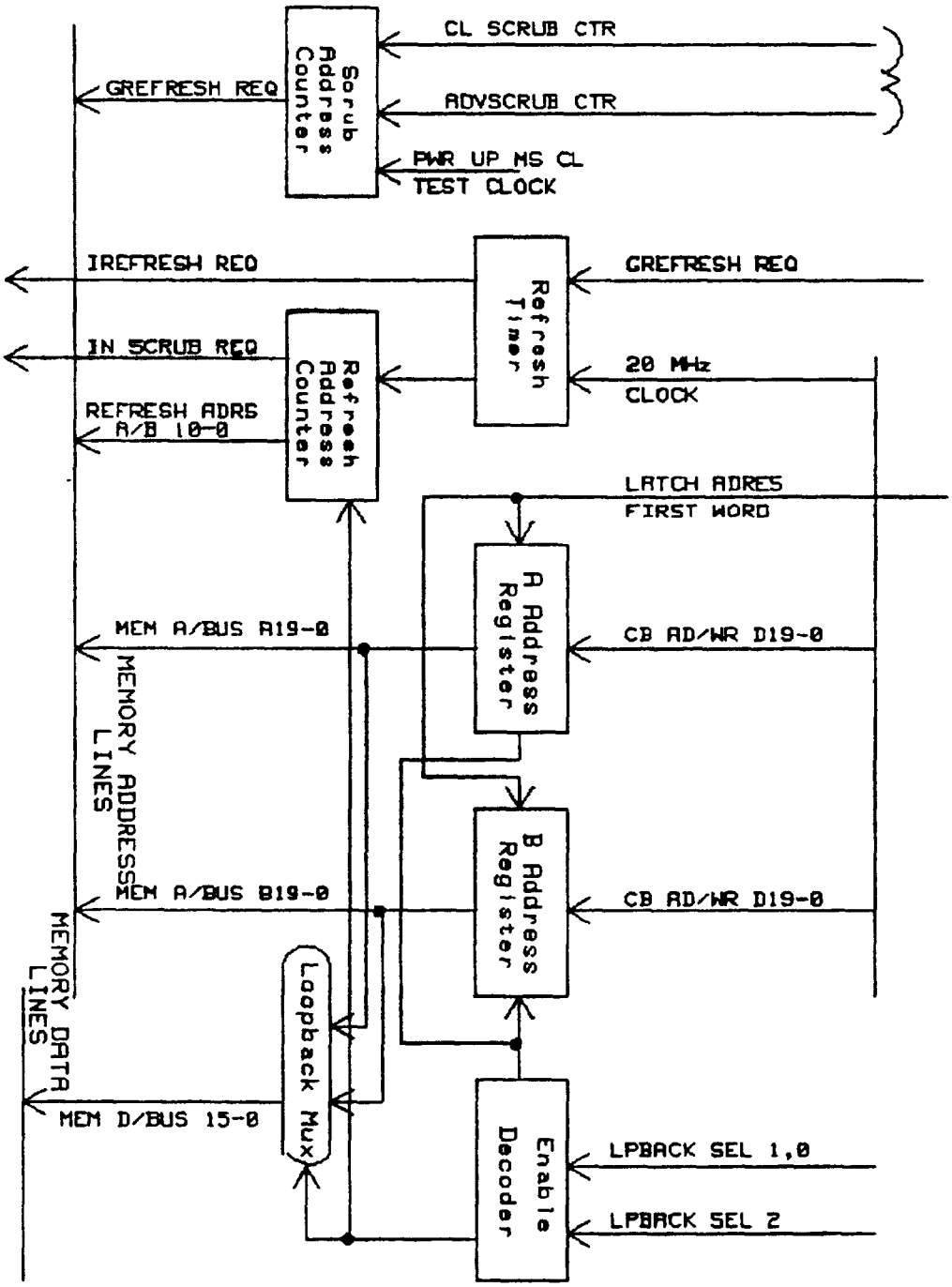



FIGURE 1. JMM functional block diagram.



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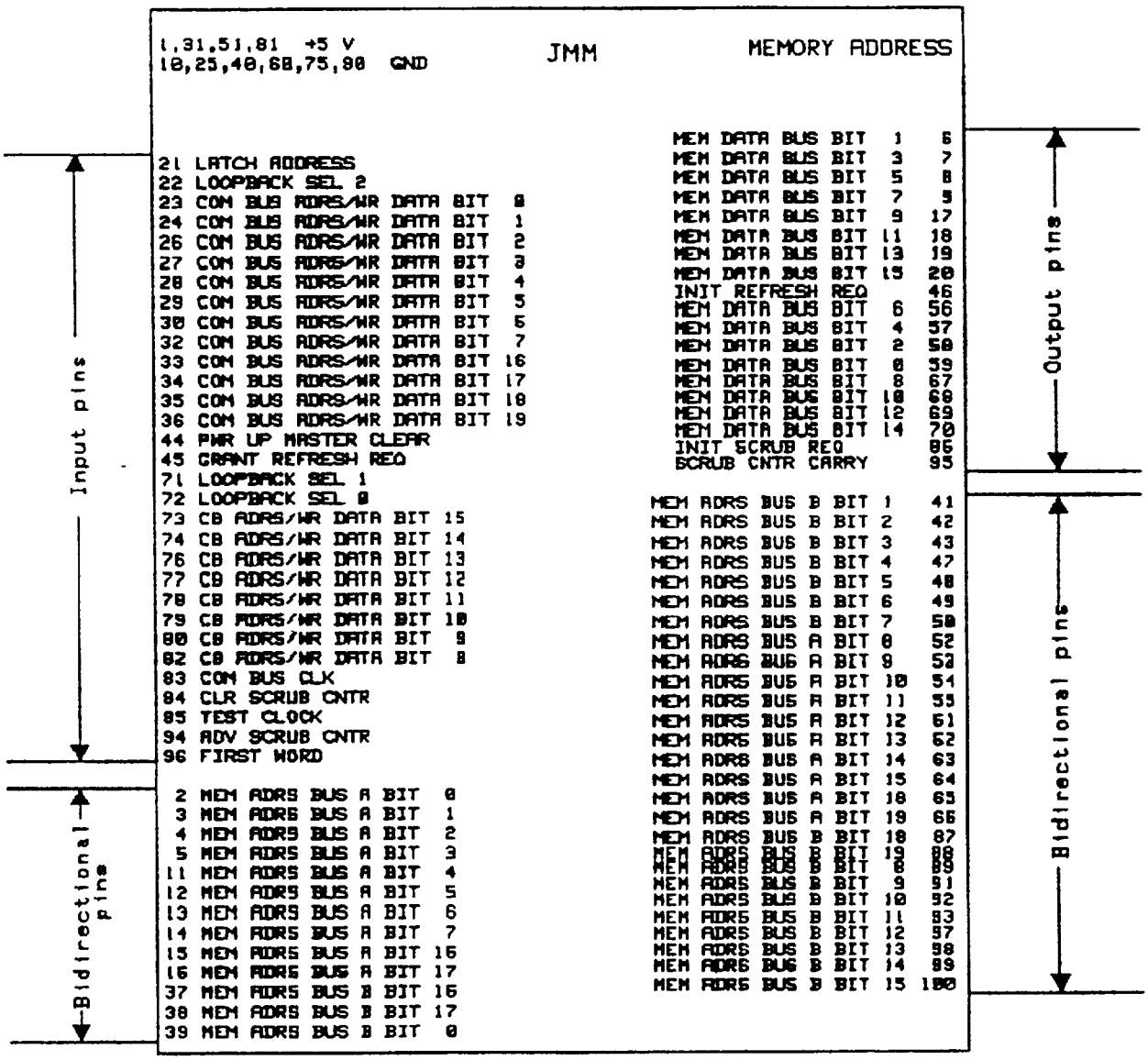



FIGURE 2. JMM pin assignment.



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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, L, and S are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E, L or S are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, L, and S where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

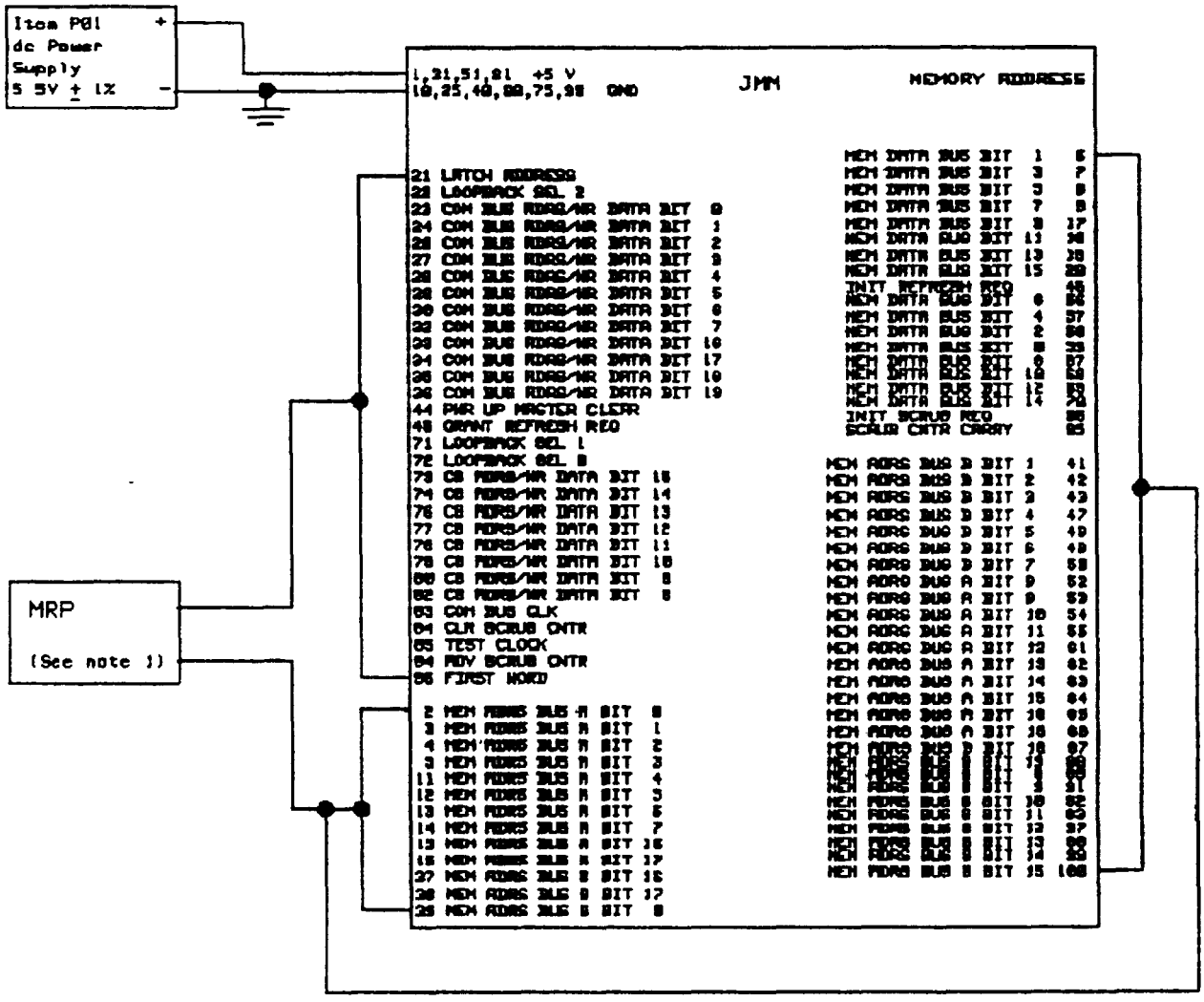
4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.



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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. JMM operating temperature and component temperature test circuit.



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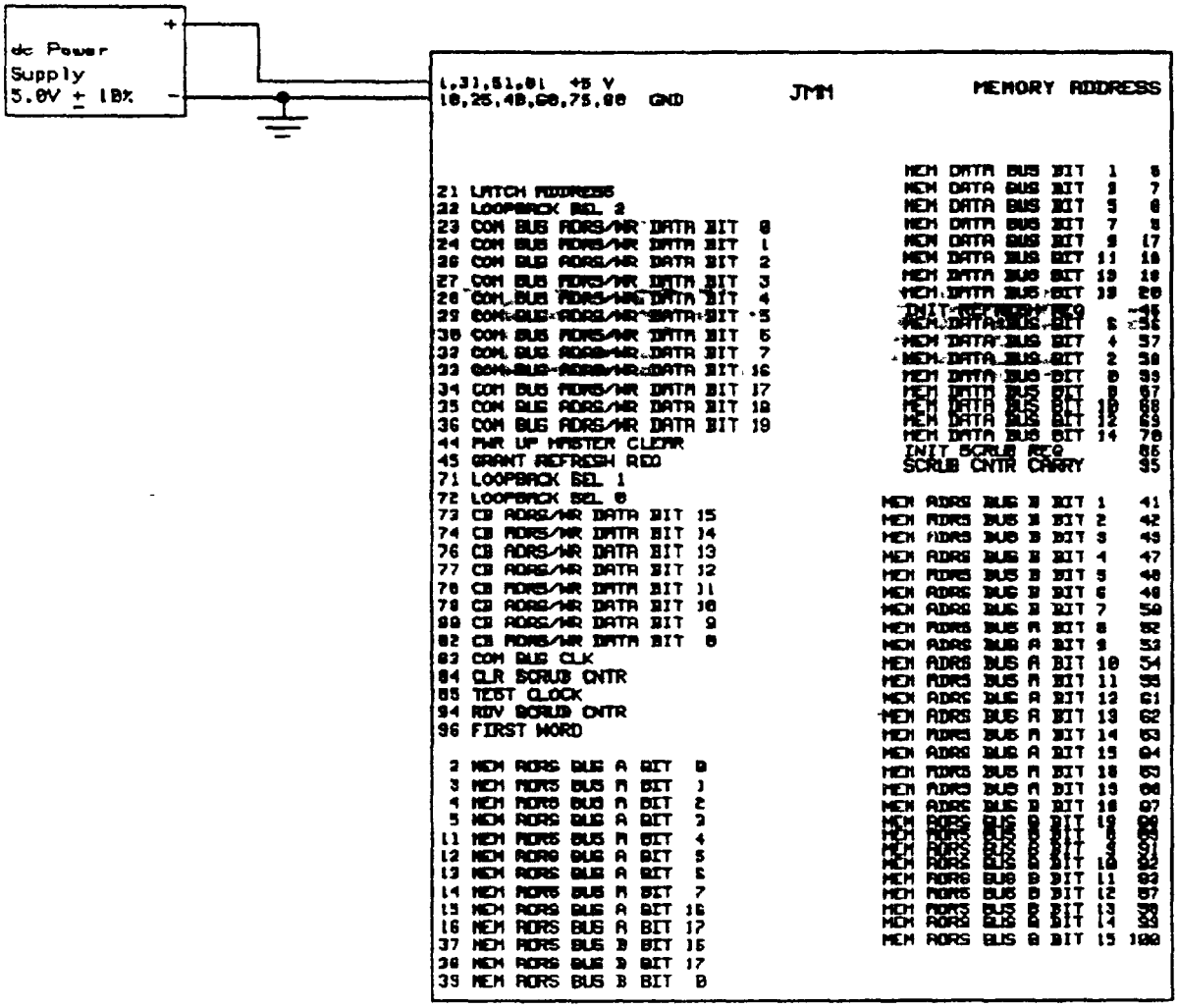



FIGURE 4. JMM vibration test circuit.



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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

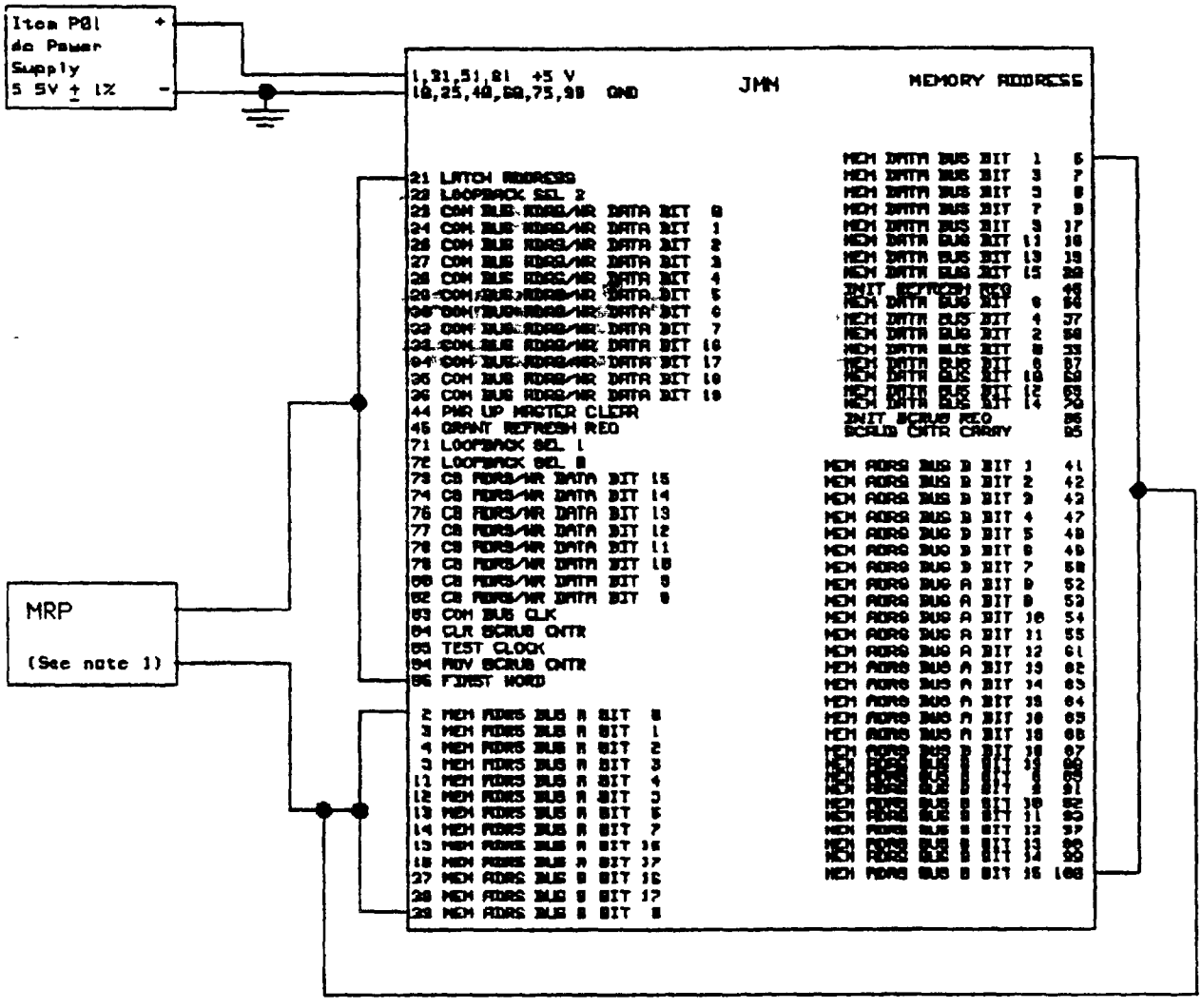
4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.



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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. JMM life test circuit.



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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 8.98 W (max); 6.02 W (typical).



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6.3.1 Input parameters.

| Signal | Vic min (V) | Vil max (V) | Vih min (V) | Iil min (mA) | Iih max (mA) | Ibd max (mA) | Cin max (pF) |
|---|---|---|---|---|---|---|---|
| LATCH ADRES | -1.2 | 0.4 | 2.4 | -4.8 | 0.16 | 0.8 | 216 |
| LPBACK SEL 2 | -1.2 | 0.4 | 2.4 | -1.8 | 0.06 | 0.3 | 100 |
| CB AD/WR D0 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 100 |
| CB AD/WR D1 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 100 |
| CB AD/WR D2 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 100 |
| CB AD/WR D3 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 100 |
| CB AD/WR D4 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 100 |
| CB AD/WR D5 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 100 |
| CB AD/WR D6 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 100 |
| CB AD/WR D7 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 100 |
| CB AD/WR D16 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 75 |
| CB AD/WR D17 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 75 |
| CB AD/WR D18 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 75 |



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6.3.1 Input parameters - Continued.

| Signal | Vic min (V) | Vil max (V) | Vih min (V) | Iil min (mA) | Iih max (mA) | Ibd max (mA) | Cin max (pF) |
|---|---|---|---|---|---|---|---|
| CB AD/WR D19 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 75 |
| PWR UP MS CL | -1.2 | 0.4 | 2.4 | -0.6 | 0.02 | 0.1 | 75 |
| GREFRESH REQ | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 100 |
| LPBACK SEL 1 | -1.2 | 0.4 | 2.4 | -4.2 | 0.14 | 0.7 | 200 |
| LPBACK SEL 0 | -1.2 | 0.4 | 2.4 | -3.0 | 0.1 | 0.5 | 125 |
| CB AD/WR D15 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 100 |
| CB AD/WR D14 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 100 |
| CB AD/WR D13 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 100 |
| CB AD/WR D12 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 100 |
| CB AD/WR D11 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 100 |
| CB AD/WR D10 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 100 |
| CB AD/WR D9 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 100 |
| CB AD/WR D8 | -1.2 | 0.4 | 2.4 | -1.2 | 0.04 | 0.2 | 100 |



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6.3.1 Input parameters - Continued.

| Signal | Vic min (V) | Vil max (V) | Vih min (V) | Iil min (mA) | Iih max (mA) | Ibd max (mA) | Cin max (pF) |
|---|---|---|---|---|---|---|---|
| COM BUS CLK | -1.2 | 0.4 | 2.4 | -0.6 | 0.02 | 0.1 | 75 |
| CL SCRUB CTR | -1.2 | 0.4 | 2.4 | -0.6 | 0.02 | 0.1 | 75 |
| TEST CLOCK | -1.2 | 0.4 | 2.4 | -3.31 | -1.41 | 0.16 | 75 |
| ADVSCRUB CTR | -1.2 | 0.4 | 2.4 | -0.6 | 0.02 | 0.1 | 75 |
| FIRST WORD | -1.2 | 0.4 | 2.4 | -1.8 | 0.06 | 0.3 | 125 |
| MEM A/BUS A0 | NA | 0.4 | 2.4 | -1.35 | 0.19 | NA | 125 |
| MEM A/BUS A1 | NA | 0.4 | 2.4 | -1.35 | 0.19 | NA | 125 |
| MEM A/BUS A2 | NA | 0.4 | 2.4 | -1.35 | 0.19 | NA | 125 |
| MEM A/BUS A3 | NA | 0.4 | 2.4 | -1.35 | 0.19 | NA | 125 |
| MEM A/BUS A4 | NA | 0.4 | 2.4 | -1.35 | 0.19 | NA | 125 |
| MEM A/BUS A5 | NA | 0.4 | 2.4 | -1.35 | 0.19 | NA | 125 |
| MEM A/BUS A6 | NA | 0.4 | 2.4 | -1.35 | 0.19 | NA | 125 |
| MEM A/BUS A7 | NA | 0.4 | 2.4 | -1.35 | 0.19 | NA | 125 |



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6.3.1 Input parameters - Continued.

| Signal | Vic min (V) | Vil max (V) | Vih min (V) | Iil min (mA) | Iih max (mA) | Ibd max (mA) | Cin max (pF) |
|---|---|---|---|---|---|---|---|
| MEM A/BUSA16 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 125 |
| MEM A/BUSA17 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 125 |
| MEM A/BUSB16 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 125 |
| MEM A/BUSB17 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 150 |
| MEM A/BUS B0 | NA | 0.4 | 2.4 | -1.35 | 0.19 | NA | 150 |
| MEM A/BUS B1 | NA | 0.4 | 2.4 | -1.35 | 0.19 | NA | 125 |
| MEM A/BUS B2 | NA | 0.4 | 2.4 | -1.35 | 0.19 | NA | 100 |
| MEM A/BUS B3 | NA | 0.4 | 2.4 | -1.35 | 0.19 | NA | 125 |
| MEM A/BUS B4 | NA | 0.4 | 2.4 | -1.35 | 0.19 | NA | 125 |
| MEM A/BUS B5 | NA | 0.4 | 2.4 | -1.35 | 0.19 | NA | 125 |
| MEM A/BUS B6 | NA | 0.4 | 2.4 | -1.35 | 0.19 | NA | 125 |
| MEM A/BUS B7 | NA | 0.4 | 2.4 | -1.35 | 0.19 | NA | 125 |
| MEM A/BUS A8 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 100 |



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6.3.1 Input parameters - Continued.

| Signal | Vic min (V) | Vil max (V) | Vih min (V) | Iil min (mA) | Iih max (mA) | Ibd max (mA) | Cin max (pF) |
|---|---|---|---|---|---|---|---|
| MEM A/BUS A9 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 100 |
| MEM A/BUSA10 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 100 |
| MEM A/BUSA11 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 100 |
| MEM A/BUSA12 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 125 |
| MEM A/BUSA13 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 100 |
| MEM A/BUSA14 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 100 |
| MEM A/BUSA15 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 100 |
| MEM A/BUSA18 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 100 |
| MEM A/BUSA19 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 150 |
| MEM A/BUSB18 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 214 |
| MEM A/BUSB19 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 75 |
| MEM A/BUS B8 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 100 |
| MEM A/BUS B9 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 125 |



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6.3.1 Input parameters - Continued.

| Signal | Vic min (V) | Vil max (V) | Vih min (V) | Iil min (mA) | Iih max (mA) | Ibd max (mA) | Cin max (pF) |
|---|---|---|---|---|---|---|---|
| MEM A/BUSB10 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 100 |
| MEM A/BUSB11 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 100 |
| MEM A/BUSB12 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 100 |
| MEM A/BUSB13 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 100 |
| MEM A/BUSB14 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 125 |
| MEM A/BUSB15 | NA | 0.4 | 2.4 | -0.75 | 0.17 | NA | 125 |



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6.3.2 Output parameters.

| Signal | Voh min (V) | Vol max (V) | Ioh max (mA) | Iol max (mA) | Iozh max (mA) | Iozl max (mA) | Ios max (mA) | Cout max (pF) |
|---|---|---|---|---|---|---|---|---|
| MEM A/BUS A0 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM A/BUS A1 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM A/BUS A2 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM A/BUS A3 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM A/BUS A4 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM A/BUS A5 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM A/BUS A6 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM A/BUS A7 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM A/BUSA16 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM A/BUSA17 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM A/BUSB16 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM A/BUSB17 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 150 |
| MEM A/BUS B0 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 150 |



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6.3.2 Output parameters - Continued.

| Signal | Voh min (V) | Vol max (V) | Ioh max (mA) | Iol max (mA) | Iozh max (mA) | Iozl max (mA) | Ios max (mA) | Cout max (pF) |
|---|---|---|---|---|---|---|---|---|
| MEM A/BUS B1 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM A/BUS B2 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 100 |
| MEM A/BUS B3 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM A/BUS B4 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM A/BUS B5 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM A/BUS B6 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM A/BUS B7 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM A/BUS A8 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 100 |
| MEM A/BUS A9 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 100 |
| MEM A/BUSA10 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 100 |
| MEM A/BUSA11 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 100 |
| MEM A/BUSA12 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM A/BUSA13 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 100 |



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6.3.2 Output parameters - Continued.

| Signal | Voh min (V) | Vol max (V) | Ioh max (mA) | Iol max (mA) | Iozh max (mA) | Iozl max (mA) | Ios max (mA) | Cout max (pF) |
|---|---|---|---|---|---|---|---|---|
| MEM A/BUSA14 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 100 |
| MEM A/BUSA15 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 100 |
| MEM A/BUSA18 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 100 |
| MEM A/BUSA19 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 150 |
| MEM A/BUSB18 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 214 |
| MEM A/BUSB19 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 75 |
| MEM A/BUS B8 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 100 |
| MEM A/BUS B9 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM A/BUSB10 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 100 |
| MEM A/BUSB11 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 100 |
| MEM A/BUSB12 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 100 |
| MEM A/BUSB13 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 100 |
| MEM A/BUSB14 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |



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6.3.2 Output parameters - Continued.

| Signal | Voh min (V) | Vol max (V) | Ioh max (mA) | Iol max (mA) | Iozh max (mA) | Iozl max (mA) | Ios max (mA) | Cout max (pF) |
|---|---|---|---|---|---|---|---|---|
| MEM A/BUSB15 | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -100.0<br>- 60.0 | 125 |
| MEM D/BUS 1 | 2.4 | 0.5 | -0.4 | 6.0 | 0.05 | 0.05 | -60.0 | 75 |
| MEM D/BUS 3 | 2.4 | 0.5 | -0.4 | 6.0 | 0.05 | 0.05 | -60.0 | 75 |
| MEM D/BUS 5 | 2.4 | 0.5 | -0.4 | 6.0 | 0.05 | 0.05 | -60.0 | 75 |
| MEM D/BUS 7 | 2.4 | 0.5 | -0.4 | 6.0 | 0.05 | 0.05 | -60.0 | 75 |
| MEM D/BUS 9 | 2.4 | 0.5 | -0.4 | 6.0 | 0.05 | 0.05 | -60.0 | 75 |
| MEM D/BUS 11 | 2.4 | 0.5 | -0.4 | 6.0 | 0.05 | 0.05 | -60.0 | 75 |
| MEM D/BUS 13 | 2.4 | 0.5 | -0.4 | 6.0 | 0.05 | 0.05 | -60.0 | 75 |
| MEM D/BUS 15 | 2.4 | 0.5 | -0.4 | 6.0 | 0.05 | 0.05 | -60.0 | 75 |
| IREFRESH REQ | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -60.0 | NA |
| MEM D/BUS 6 | 2.4 | 0.5 | -0.4 | 6.0 | 0.05 | 0.05 | -60.0 | 75 |
| MEM D/BUS 4 | 2.4 | 0.5 | -0.4 | 6.0 | 0.05 | 0.05 | -60.0 | 75 |
| MEM D/BUS 2 | 2.4 | 0.5 | -0.4 | 6.0 | 0.05 | 0.05 | -60.0 | 75 |



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6.3.2 Output parameters - Continued.

| Signal | Voh min (V) | Vol max (V) | Ioh max (mA) | Iol max (mA) | Iozh max (mA) | Iozl max (mA) | Ios max (mA) | Cout max (pF) |
|---|---|---|---|---|---|---|---|---|
| MEM D/BUS 0 | 2.4 | 0.5 | -0.4 | 6.0 | 0.05 | 0.05 | -60.0 | 75 |
| MEM D/BUS 8 | 2.4 | 0.5 | -0.4 | 6.0 | 0.05 | 0.05 | -60.0 | 75 |
| MEM D/BUS 10 | 2.4 | 0.5 | -0.4 | 6.0 | 0.05 | 0.05 | -60.0 | 75 |
| MEM D/BUS 12 | 2.4 | 0.5 | -0.4 | 6.0 | 0.05 | 0.05 | -60.0 | 75 |
| MEM D/BUS 14 | 2.4 | 0.5 | -0.4 | 6.0 | 0.05 | 0.05 | -60.0 | 75 |
| IN SCRUB REQ | 2.4 | 0.5 | -0.4 | 6.0 | NA | NA | -60.0 | NA |
| SCRUB CTR CA | 2.4 | 0.5 | -0.16 | 1.6 | NA | NA | -20.0 | NA |



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6.3.3 Class I and II dynamic parameters.

| Ref num | Sym | Parameter | Minimum limit | | | Maximum limit | | | Unit | Figs |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | -55°C | 25°C | 85°C | -55°C | 25°C | 85°C | | |
| 1 | tpd | LPBACK SEL 0,1 to MEM D/BUS 7-4 | - | - | - | 22 | 22 | 22 | nS | 11 |
| 2 | tpd | LPBACK SEL 2 to MEM D/BUS 15-0 | - | - | - | 32 | 32 | 32 | nS | 11 |
| 3 | tpd | GREFRESH REQ to IREFRESH REQ | - | - | - | 41 | 41 | 41 | nS | 11 |
| 4 | tpd | GREFRESH REQ to IN SCRUB REQ | - | - | - | 52 | 52 | 52 | nS | 11 |
| 5 | tpd | LPBACK SEL 1 to MEM A/BUS A19-0 | - | - | - | 51 | 51 | 51 | nS | 11 |
| 5 | tpd | LPBACK SEL 1 to MEM A/BUS B19-0 | - | - | - | 51 | 51 | 51 | nS | 11 |
| 6 | tpd | FIRST WORD to MEM A/BUS A15-0 | - | - | - | 36 | 36 | 36 | nS | 11 |
| 6 | tpd | FIRST WORD to MEM A/BUS B15-0 | - | - | - | 36 | 36 | 36 | nS | 11 |



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6.3.3 Class I and II dynamic parameters - Continued.

| Ref num | Sym | Parameter | Minimum limit | | | Maximum limit | | | Unit | Figs |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | -55°C | 25°C | 85°C | -55°C | 25°C | 85°C | | |
| 7 | tpd | LPBACK SEL 0 to MEM A/BUS A19-8 | - | - | - | 51 | 51 | 51 | nS | 11 |
| 7 | tpd | LPBACK SEL 0 to MEM A/BUS B19-8 | - | - | - | 51 | 51 | 51 | nS | 11 |
| 8 | tpd | LATCH ADRES to MEM A/BUS A19-0 | - | - | - | 25 | 25 | 25 | nS | 11 |
| 8 | tpd | LATCH ADRES to MEM A/BUS B19-0 | - | - | - | 25 | 25 | 25 | nS | 11 |
| 9 | tpd | CB AD/WR D19-0 to MEM A/BUS A19-0 | - | - | - | 22 | 22 | 22 | nS | 11 |
| 9 | tpd | CB AD/WR D19-0 to MEM A/BUS B19-0 | - | - | - | 22 | 22 | 22 | nS | 11 |



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6.4 Module function. The memory address module contains addressing logic for the memory array modules, refresh logic for the Dynamic Random Access Memories (DRAMs), memory error scrub generation circuits and loopback logic. The address information is taken from the common bus and latched into the address registers which drive the memory address lines. As DRAM design is based on capacitor charge storage for each bit in the array, the refresh counter and timer is used to generate refresh signals to offset data degradation caused by time and temperature. The refresh counter and timer generate refresh address information once every 15.6 microseconds. A normal read or write to a DRAM serves to refresh all the bits associated with the particular row decoded. When enabled by the memory timing module, the scrub counter and timer generate a scrub address every 1.0 mS. The loopback logic is used during Built-In-Test (BIT) tests to verify the integrity of memory address and data paths.



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6.4.3 Description of I/O pins.

| Signal name | Pin number | I/O | Description |
|---|---|---|---|
| LATCH ADRES | 21 | I | LATCH ADDRESS |
| LPBACK SEL 2 | 22 | I | LOOPBACK SEL 2 |
| CB AD/WR D0 | 23 | I | COM BUS ADRS/WR DATA BIT 0 |
| CB AD/WR D1 | 24 | I | COM BUS ADRS/WR DATA BIT 1 |
| CB AD/WR D2 | 26 | I | COM BUS ADRS/WR DATA BIT 2 |
| CB AD/WR D3 | 27 | I | COM BUS ADRS/WR DATA BIT 3 |
| CB AD/WR D4 | 28 | I | COM BUS ADRS/WR DATA BIT 4 |
| CB AD/WR D5 | 29 | I | COM BUS ADRS/WR DATA BIT 5 |
| CB AD/WR D6 | 30 | I | COM BUS ADRS/WR DATA BIT 6 |
| CB AD/WR D7 | 32 | I | COM BUS ADRS/WR DATA BIT 7 |
| CB AD/WR D16 | 33 | I | COM BUS ADRS/WR DATA BIT 16 |
| CB AD/WR D17 | 34 | I | COM BUS ADRS/WR DATA BIT 17 |
| CB AD/WR D18 | 35 | I | COM BUS ADRS/WR DATA BIT 18 |
| CB AD/WR D19 | 36 | I | COM BUS ADRS/WR DATA BIT 19 |
| PWR UP MS CL | 44 | I | PWR UP MASTER CLEAR |
| GREFRESH REQ | 45 | I | GRANT REFRESH REQ |
| LPBACK SEL 1 | 71 | I | LOOPBACK SEL 1 |
| LPBACK SEL 0 | 72 | I | LOOPBACK SEL 0 |
| CB AD/WR D15 | 73 | I | COM BUS ADRS/WR DATA BIT 15 |
| CB AD/WR D14 | 74 | I | COM BUS ADRS/WR DATA BIT 14 |



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6.4.3 Description of I/O pins - Continued.

| Signal name | Pin number | I/O | Description |
|---|---|---|---|
| CB AD/WR D13 | 76 | I | COM BUS ADRS/WR DATA BIT 13 |
| CB AD/WR D12 | 77 | I | COM BUS ADRS/WR DATA BIT 12 |
| CB AD/WR D11 | 78 | I | COM BUS ADRS/WR DATA BIT 11 |
| CB AD/WR D10 | 79 | I | COM BUS ADRS/WR DATA BIT 10 |
| CB AD/WR D9 | 80 | I | COM BUS ADRS/WR DATA BIT 9 |
| CB AD/WR D8 | 82 | I | COM BUS ADRS/WR DATA BIT 8 |
| COM BUS CLK | 83 | I | COM BUS CLK |
| CL SCRUB CTR | 84 | I | CLR SCRUB CNTR |
| TEST CLOCK | 85 | I | TEST CLOCK |
| ADVSCRUB CTR | 94 | I | ADV SCRUB CNTR |
| FIRST WORD | 96 | I | FIRST WORD |
| MEM A/BUS A0 | 2 | I/O | MEM ADRS BUS A BIT 0 |
| MEM A/BUS A1 | 3 | I/O | MEM ADRS BUS A BIT 1 |
| MEM A/BUS A2 | 4 | I/O | MEM ADRS BUS A BIT 2 |
| MEM A/BUS A3 | 5 | I/O | MEM ADRS BUS A BIT 3 |
| MEM A/BUS A4 | 11 | I/O | MEM ADRS BUS A BIT 4 |
| MEM A/BUS A5 | 12 | I/O | MEM ADRS BUS A BIT 5 |
| MEM A/BUS A6 | 13 | I/O | MEM ADRS BUS A BIT 6 |
| MEM A/BUS A7 | 14 | I/O | MEM ADRS BUS A BIT 7 |
| MEM A/BUSA16 | 15 | I/O | MEM ADRS BUS A BIT 16 |



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6.4.3 Description of I/O pins - Continued.

| Signal name | Pin number | I/O | Description |
|---|---|---|---|
| MEM A/BUSA17 | 16 | I/O | MEM ADRS BUS A BIT 17 |
| MEM A/BUSB16 | 37 | I/O | MEM ADRS BUS B BIT 16 |
| MEM A/BUSB17 | 38 | I/O | MEM ADRS BUS B BIT 17 |
| MEM A/BUS B0 | 39 | I/O | MEM ADRS BUS B BIT 0 |
| MEM A/BUS B1 | 41 | I/O | MEM ADRS BUS B BIT 1 |
| MEM A/BUS B2 | 42 | I/O | MEM ADRS BUS B BIT 2 |
| MEM A/BUS B3 | 43 | I/O | MEM ADRS BUS B BIT 3 |
| MEM A/BUS B4 | 47 | I/O | MEM ADRS BUS B BIT 4 |
| MEM A/BUS B5 | 48 | I/O | MEM ADRS BUS B BIT 5 |
| MEM A/BUS B6 | 49 | I/O | MEM ADRS BUS B BIT 6 |
| MEM A/BUS B7 | 50 | I/O | MEM ADRS BUS B BIT 7 |
| MEM A/BUS A8 | 52 | I/O | MEM ADRS BUS A BIT 8 |
| MEM A/BUS A9 | 53 | I/O | MEM ADRS BUS A BIT 9 |
| MEM A/BUSA10 | 54 | I/O | MEM ADRS BUS A BIT 10 |
| MEM A/BUSA11 | 55 | I/O | MEM ADRS BUS A BIT 11 |
| MEM A/BUSA12 | 61 | I/O | MEM ADRS BUS A BIT 12 |
| MEM A/BUSA13 | 62 | I/O | MEM ADRS BUS A BIT 13 |
| MEM A/BUSA14 | 63 | I/O | MEM ADRS BUS A BIT 14 |
| MEM A/BUSA15 | 64 | I/O | MEM ADRS BUS A BIT 15 |
| MEM A/BUSA18 | 65 | I/O | MEM ADRS BUS A BIT 18 |



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6.4.3 Description of I/O pins - Continued.

| Signal name | Pin number | I/O | Description |
|---|---|---|---|
| MEM A/BUSA19 | 66 | I/O | MEM ADRS BUS A BIT 19 |
| MEM A/BUSB18 | 87 | I/O | MEM ADRS BUS B BIT 18 |
| MEM A/BUSB19 | 88 | I/O | MEM ADRS BUS B BIT 19 |
| MEM A/BUS B8 | 89 | I/O | MEM ADRS BUS B BIT 8 |
| MEM A/BUS B9 | 91 | I/O | MEM ADRS BUS B BIT 9 |
| MEM A/BUSB10 | 92 | I/O | MEM ADRS BUS B BIT 10 |
| MEM A/BUSB11 | 93 | I/O | MEM ADRS BUS B BIT 11 |
| MEM A/BUSB12 | 97 | I/O | MEM ADRS BUS B BIT 12 |
| MEM A/BUSB13 | 98 | I/O | MEM ADRS BUS B BIT 13 |
| MEM A/BUSB14 | 99 | I/O | MEM ADRS BUS B BIT 14 |
| MEM A/BUSB15 | 100 | I/O | MEM ADRS BUS B BIT 15 |
| MEM D/BUS 1 | 6 | O | MEM DATA BUS BIT 1 |
| MEM D/BUS 3 | 7 | O | MEM DATA BUS BIT 3 |
| MEM D/BUS 5 | 8 | O | MEM DATA BUS BIT 5 |
| MEM D/BUS 7 | 9 | O | MEM DATA BUS BIT 7 |
| MEM D/BUS 9 | 17 | O | MEM DATA BUS BIT 9 |
| MEM D/BUS 11 | 18 | O | MEM DATA BUS BIT 11 |
| MEM D/BUS 13 | 19 | O | MEM DATA BUS BIT 13 |
| MEM D/BUS 15 | 20 | O | MEM DATA BUS BIT 15 |
| IREFRESH REQ | 46 | O | INIT REFRESH REQ |



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6.4.3 Description of I/O pins - Continued.

| Signal name | Pin number | I/O | Description |
|---|---|---|---|
| MEM D/BUS 6 | 56 | O | MEM DATA BUS BIT 6 |
| MEM D/BUS 4 | 57 | O | MEM DATA BUS BIT 4 |
| MEM D/BUS 2 | 58 | O | MEM DATA BUS BIT 2 |
| MEM D/BUS 0 | 59 | O | MEM DATA BUS BIT 0 |
| MEM D/BUS 8 | 67 | O | MEM DATA BUS BIT 8 |
| MEM D/BUS 10 | 68 | O | MEM DATA BUS BIT 10 |
| MEM D/BUS 12 | 69 | O | MEM DATA BUS BIT 12 |
| MEM D/BUS 14 | 70 | O | MEM DATA BUS BIT 14 |
| IN SCRUB REQ | 86 | O | INIT SCRUB REQ |
| SCRUB CTR CA | 95 | O | SCRUB CNTR CARRY |



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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-0077)

Agent: NW



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TABLE II. Isolation test.

| Test cond | Test description | Connect | | 25 ± 5°C Initial limits | | Units | Notes |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| A | Isolation, frame | C | Frame | 10 | INF | MOhm | 1-8 |



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NOTES:

1. Isolation test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Megohm bridge voltage | 100 V | ± 10 percent | A |
| Isolation measurement | NA | ± 500 KOhm | A |

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.



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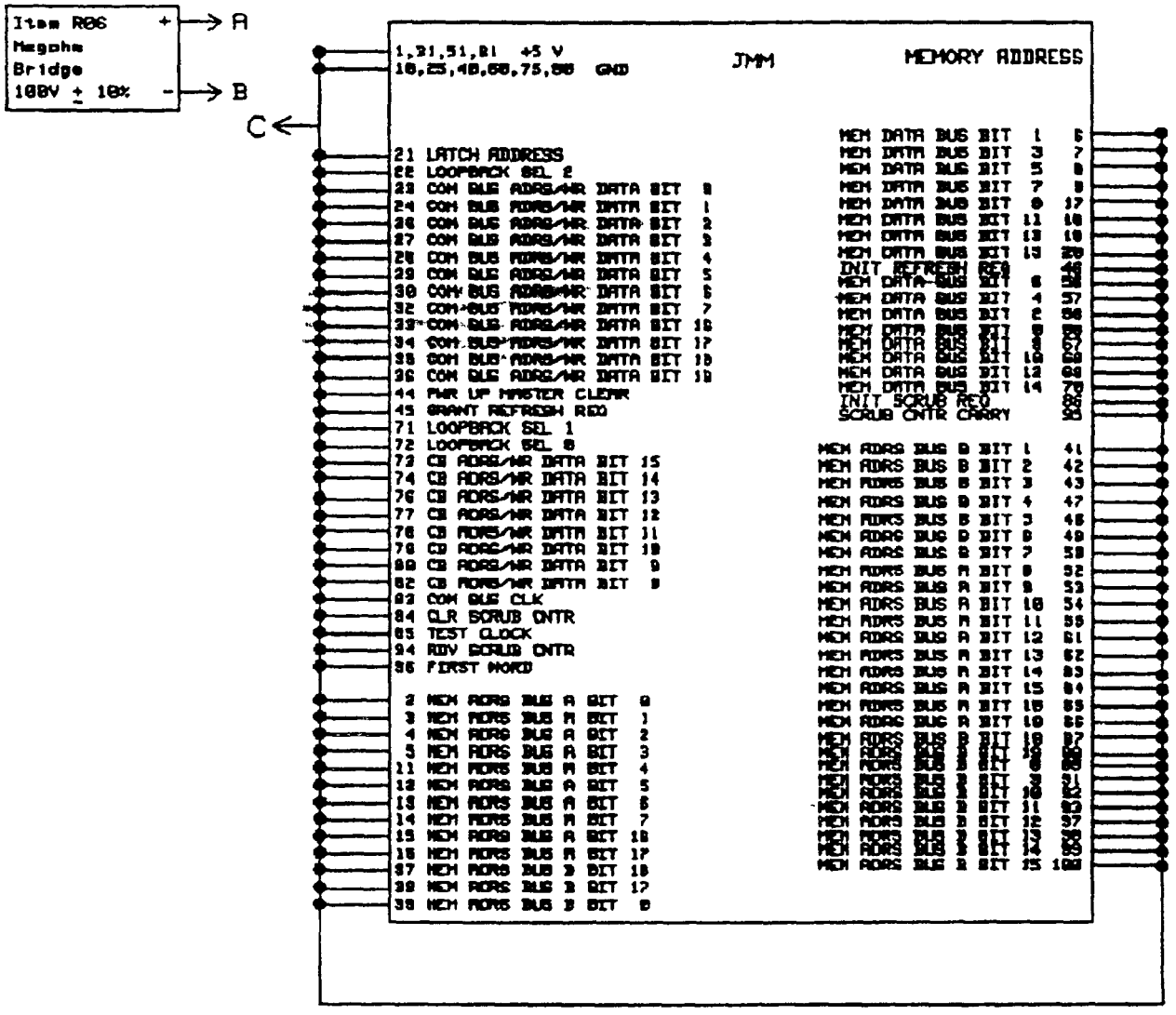



FIGURE 6. Isolation test circuit for table II.



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TABLE III. Continuity test.

| Test cond | Test description | Connect A to pin | Connect B to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| B1 | Continuity, GND | 10 | 25 | 0 | 1 | Ohm | 1-5 |
| B2 | Continuity, GND | 10 | 40 | | | | |
| B3 | Continuity, GND | 60 | 75 | | | | |
| B4 | Continuity, GND | 60 | 90 | | | | |
| B5 | Continuity, Vcc | 1 | 31 | | | | |
| B6 | Continuity, Vcc | 51 | 81 | 0 | 1 | Ohm | 1-5 |

NOTES:

1. Continuity test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Continuity measurement | NA | ± 0.05 Ohm | B1-B6 |

2. The ohmmeter used shall be item R07.

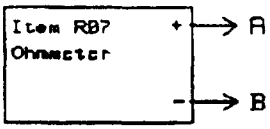
3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.



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1,31,51,81 +5 V
10,25,40,60,75,90 GND

JMM

MEMORY ADDRESS

| | | | |
|---|---|---|---|
| 21 | LATCH ADDRESS | MEM DATA BUS BIT 1 | 0 |
| 22 | LOOPBACK SEL 2 | MEM DATA BUS BIT 3 | 7 |
| 23 | COM BUS ADRS/WR DATA BIT 0 | MEM DATA BUS BIT 5 | 8 |
| 24 | COM BUS ADRS/WR DATA BIT 1 | MEM DATA BUS BIT 7 | 9 |
| 26 | COM BUS ADRS/WR DATA BIT 2 | MEM DATA BUS BIT 9 | 17 |
| 27 | COM BUS ADRS/WR DATA BIT 3 | MEM DATA BUS BIT 11 | 18 |
| 28 | COM BUS ADRS/WR DATA BIT 4 | MEM DATA BUS BIT 13 | 19 |
| 29 | COM BUS ADRS/WR DATA BIT 5 | MEM DATA BUS BIT 15 | 20 |
| 30 | COM BUS ADRS/WR DATA BIT 6 | INIT REFRESH REQ | 45 |
| 32 | COM BUS ADRS/WR DATA BIT 7 | MEM DATA BUS BIT 0 | 55 |
| 33 | COM BUS ADRS/WR DATA BIT 16 | MEM DATA BUS BIT 4 | 57 |
| 34 | COM BUS ADRS/WR DATA BIT 17 | MEM DATA BUS BIT 2 | 58 |
| 35 | COM BUS ADRS/WR DATA BIT 18 | MEM DATA BUS BIT 6 | 59 |
| 36 | COM BUS ADRS/WR DATA BIT 19 | MEM DATA BUS BIT 8 | 67 |
| 44 | PWR UP MASTER CLEAR | MEM DATA BUS BIT 10 | 69 |
| 45 | GRANT REFRESH REQ | MEM DATA BUS BIT 12 | 65 |
| 71 | LOOPBACK SEL 1 | MEM DATA BUS BIT 14 | 70 |
| 72 | LOOPBACK SEL 0 | INIT SCRUB REQ | 85 |
| 73 | CB ADRS/WR DATA BIT 15 | SCRUB CNTR CARRY | 95 |
| 74 | CB ADRS/WR DATA BIT 14 | | |
| 76 | CB ADRS/WR DATA BIT 13 | MEM ADRS BUS B BIT 1 | 41 |
| 77 | CB ADRS/WR DATA BIT 12 | MEM ADRS BUS B BIT 2 | 42 |
| 78 | CB ADRS/WR DATA BIT 11 | MEM ADRS BUS B BIT 3 | 43 |
| 79 | CB ADRS/WR DATA BIT 10 | MEM ADRS BUS B BIT 4 | 47 |
| 80 | CB ADRS/WR DATA BIT 9 | MEM ADRS BUS B BIT 5 | 48 |
| 82 | CB ADRS/WR DATA BIT 8 | MEM ADRS BUS B BIT 6 | 49 |
| 83 | COM BUS CLK | MEM ADRS BUS B BIT 7 | 50 |
| 84 | CLR SCRUB CNTR | MEM ADRS BUS A BIT 8 | 32 |
| 85 | TEST CLOCK | MEM ADRS BUS A BIT 9 | 33 |
| 94 | ADV SCRUB CNTR | MEM ADRS BUS A BIT 10 | 34 |
| 96 | FIRST WORD | MEM ADRS BUS A BIT 11 | 35 |
| | | MEM ADRS BUS A BIT 12 | 61 |
| 2 | MEM ADRS BUS A BIT 0 | MEM ADRS BUS A BIT 13 | 62 |
| 3 | MEM ADRS BUS A BIT 1 | MEM ADRS BUS A BIT 14 | 63 |
| 4 | MEM ADRS BUS A BIT 2 | MEM ADRS BUS A BIT 15 | 64 |
| 5 | MEM ADRS BUS A BIT 3 | MEM ADRS BUS A BIT 16 | 65 |
| 11 | MEM ADRS BUS A BIT 4 | MEM ADRS BUS A BIT 17 | 66 |
| 12 | MEM ADRS BUS A BIT 5 | MEM ADRS BUS B BIT 16 | 87 |
| 13 | MEM ADRS BUS A BIT 6 | MEM ADRS BUS B BIT 17 | 88 |
| 14 | MEM ADRS BUS A BIT 7 | MEM ADRS BUS B BIT 0 | 89 |
| 15 | MEM ADRS BUS A BIT 16 | MEM ADRS BUS B BIT 9 | 81 |
| 16 | MEM ADRS BUS A BIT 17 | MEM ADRS BUS B BIT 10 | 82 |
| 27 | MEM ADRS BUS B BIT 16 | MEM ADRS BUS B BIT 11 | 83 |
| 38 | MEM ADRS BUS B BIT 17 | MEM ADRS BUS B BIT 12 | 87 |
| 39 | MEM ADRS BUS B BIT 0 | MEM ADRS BUS B BIT 13 | 98 |
| | | MEM ADRS BUS B BIT 14 | 99 |
| | | MEM ADRS BUS B BIT 15 | 100 |

FIGURE 7. Continuity test circuit for table III.



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TABLE IV. Power supply stress test.

| Test cond | Power supply | Vin (V) | Connect | | 25 ± 5°C Initial limits | Notes |
|---|---|---|---|---|---|---|
| | | | Positive lead to | Negative lead to | | |
| C | Vcc | 7.0 | 1,31,51, 81 | 10,25, 40,60, 75,90 | Module shall pass all remaining tests. | 1-6 |

NOTES:

1. Power supply stress test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 7.0 V | +0 mV,-70 mV | C |
| Power supply current limit | 2.939 A | NA | C |

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.



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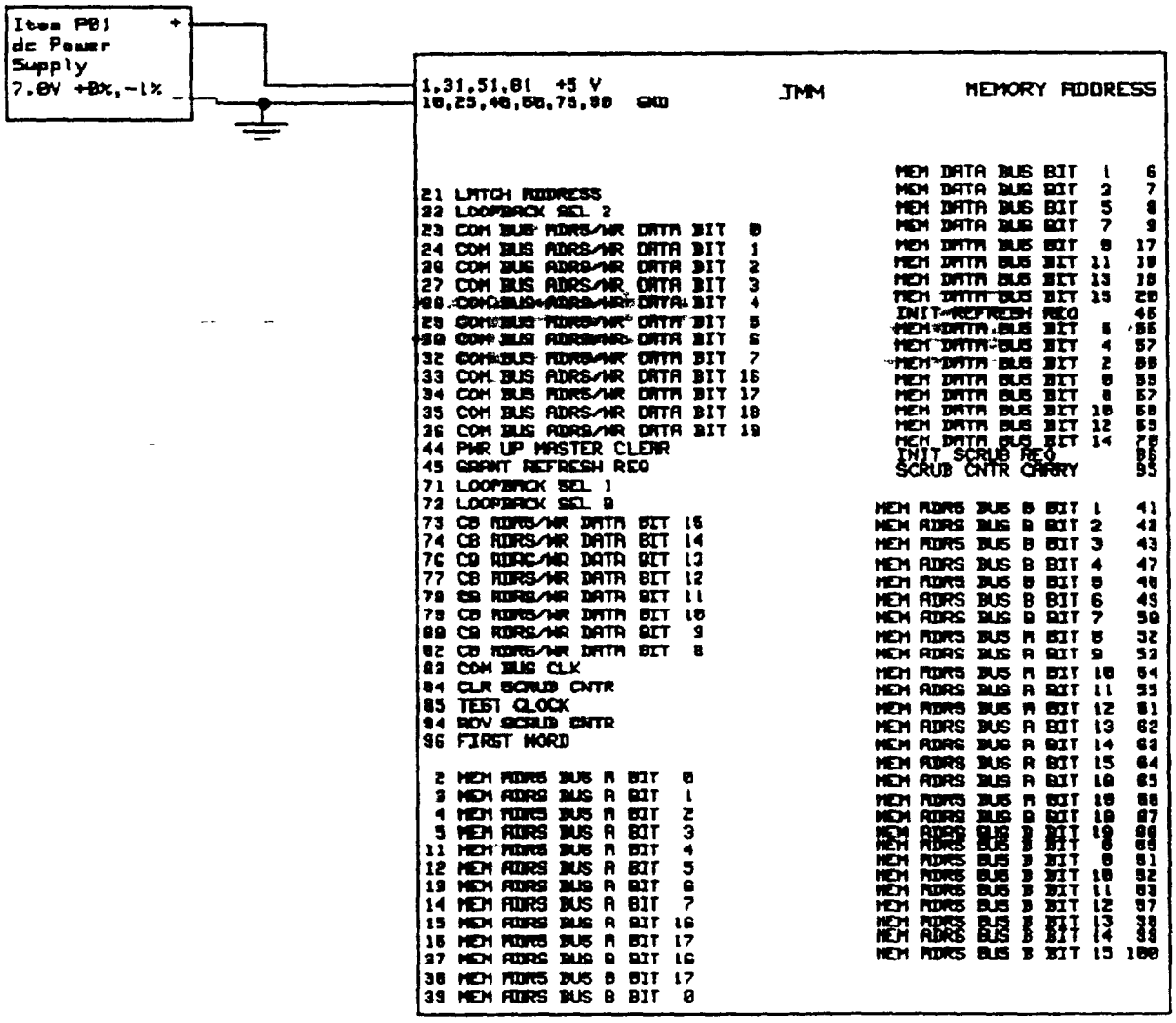



FIGURE 8. Vcc stress test circuit for table IV.



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TABLE V. Input breakdown current test.

| Test cond | Signal name | Pin no. | Vin (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| D1 | LATCH ADRES | 21 | 5.5 | 0.0 | 0.8 | mA | 1-3 |
| D2 | LPBACK SEL 2 | 22 | 5.5 | 0.0 | 0.3 | mA | 1-3 |
| D3 | CB AD/WR D0 | 23 | 5.5 | 0.0 | 0.2 | mA | 1-3 |
| D4 | CB AD/WR D1 | 24 | | | | | |
| D5 | CB AD/WR D2 | 26 | | | | | |
| D6 | CB AD/WR D3 | 27 | | | | | |
| D7 | CB AD/WR D4 | 28 | | | | | |
| D8 | CB AD/WR D5 | 29 | | | | | |
| D9 | CB AD/WR D6 | 30 | | | | | |
| D10 | CB AD/WR D7 | 32 | | | | | |
| D11 | CB AD/WR D16 | 33 | | | | | |
| D12 | CB AD/WR D17 | 34 | | | | | |
| D13 | CB AD/WR D18 | 35 | | | | | |
| D14 | CB AD/WR D19 | 36 | 5.5 | 0.0 | 0.2 | mA | 1-3 |
| D15 | PWR UP MS CL | 44 | 5.5 | 0.0 | 0.1 | mA | 1-3 |
| D16 | GREFRESH REQ | 45 | 5.5 | 0.0 | 0.2 | mA | 1-3 |
| D17 | LPBACK SEL 1 | 71 | 5.5 | 0.0 | 0.7 | mA | 1-3 |
| D18 | LPBACK SEL 0 | 72 | 5.5 | 0.0 | 0.5 | mA | 1-3 |
| D19 | CB AD/WR D15 | 73 | 5.5 | 0.0 | 0.2 | mA | 1-3 |
| D20 | CB AD/WR D14 | 74 | 5.5 | 0.0 | 0.2 | mA | 1-3 |



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TABLE V. Input breakdown current test - Continued.

| Test cond | Signal name | Pin no. | Vin (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| D21 | CB AD/WR D13 | 76 | 5.5 | 0.0 | 0.2 | mA | 1-3 |
| D22 | CB AD/WR D12 | 77 | | | | | |
| D23 | CB AD/WR D11 | 78 | | | | | |
| D24 | CB AD/WR D10 | 79 | | | | | |
| D25 | CB AD/WR D9 | 80 | | | | | |
| D26 | CB AD/WR D8 | 82 | 5.5 | 0.0 | 0.2 | mA | 1-3 |
| D27 | COM BUS CLK | 83 | 5.5 | 0.0 | 0.1 | mA | 1-3 |
| D28 | CL SCRUB CTR | 84 | 5.5 | 0.0 | 0.1 | mA | 1-3 |
| D29 | TEST CLOCK | 85 | 5.5 | -0.057 | 0.16 | mA | 1-3 |
| D30 | ADVSCRUB CTR | 94 | 5.5 | 0.0 | 0.1 | mA | 1-3 |
| D31 | FIRST WORD | 96 | 5.5 | 0.0 | 0.3 | mA | 1-3 |



Source: https://assist.dla.mil -- Downloaded: 202640-20T18:16Z
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NOTES:

1. Input breakdown current test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 5.5 V | ± 55 mV | D1-D31 |
| Power supply current limit | 2.45 A | NA | D1-D31 |
| Forcing voltage, (Vin) | 5.5 V | ± 55 mV | D1-D31 |
| Pattern voltage | | | |
| high, (VH) | 3.0 V | ± 100 mV | D1-D31 |
| low, (VL) | 0.0 V | ± 100 mV | D1-D31 |
| Input current measurement | NA | ± 40.0 uA | D1 |
| | NA | ± 15.0 uA | D2 |
| | NA | ± 10.0 uA | D3-D14 |
| | NA | ± 5.0 uA | D15 |
| | NA | ± 10.0 uA | D16 |
| | NA | ± 35.0 uA | D17 |
| | NA | ± 25.0 uA | D18 |
| | NA | ± 10.0 uA | D19-D26 |
| | NA | ± 5.0 uA | D27,D28 |
| | NA | ± 10.8 uA | D29 |
| | NA | ± 5.0 uA | D30 |
| | NA | ± 15.0 uA | D31 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.



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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.



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TABLE VI. Output short circuit current test.

| Test cond | Signal name | Pin no. | Vout (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| E1 | MEM A/BUS A0 | 2 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E2 | MEM A/BUS A0 | 2 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E3 | MEM A/BUS A0 | 2 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E4 | MEM A/BUS A1 | 3 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E5 | MEM A/BUS A1 | 3 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E6 | MEM A/BUS A1 | 3 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E7 | MEM A/BUS A2 | 4 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E8 | MEM A/BUS A2 | 4 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E9 | MEM A/BUS A2 | 4 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E10 | MEM A/BUS A3 | 5 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E11 | MEM A/BUS A3 | 5 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E12 | MEM A/BUS A3 | 5 | | | | | |
| E13 | MEM D/BUS 1 | 6 | | | | | |
| E14 | MEM D/BUS 3 | 7 | | | | | |
| E15 | MEM D/BUS 5 | 8 | | | | | |
| E16 | MEM D/BUS 7 | 9 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E17 | MEM A/BUS A4 | 11 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E18 | MEM A/BUS A4 | 11 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E19 | MEM A/BUS A4 | 11 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E20 | MEM A/BUS A5 | 12 | 0.0 | -325.0 | -100.0 | mA | 1-4 |



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TABLE VI. Output short circuit current test - Continued.

| Test cond | Signal name | Pin no. | Vout (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| E21 | MEM A/BUS A5 | 12 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E22 | MEM A/BUS A5 | 12 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E23 | MEM A/BUS A6 | 13 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E24 | MEM A/BUS A6 | 13 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E25 | MEM A/BUS A6 | 13 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E26 | MEM A/BUS A7 | 14 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E27 | MEM A/BUS A7 | 14 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E28 | MEM A/BUS A7 | 14 | | | | | |
| E29 | MEM A/BUSA16 | 15 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E30 | MEM A/BUSA16 | 15 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E31 | MEM A/BUSA16 | 15 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E32 | MEM A/BUSA17 | 16 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E33 | MEM A/BUSA17 | 16 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E34 | MEM A/BUSA17 | 16 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E35 | MEM D/BUS 9 | 17 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E36 | MEM D/BUS 11 | 18 | | | | | |
| E37 | MEM D/BUS 13 | 19 | | | | | |
| E38 | MEM D/BUS 15 | 20 | | | | | |
| E39 | MEM A/BUSB16 | 37 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E40 | MEM A/BUSB16 | 37 | 0.0 | -325.0 | -100.0 | mA | 1-4 |



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TABLE VI. Output short circuit current test - Continued.

| Test cond | Signal name | Pin no. | Vout (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| E41 | MEM A/BUSB16 | 37 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E42 | MEM A/BUSB17 | 38 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E43 | MEM A/BUSB17 | 38 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E44 | MEM A/BUSB17 | 38 | | | | | |
| E45 | MEM A/BUS B0 | 39 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E46 | MEM A/BUS B0 | 39 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E47 | MEM A/BUS B0 | 39 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E48 | MEM A/BUS B1 | 41 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E49 | MEM A/BUS B1 | 41 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E50 | MEM A/BUS B1 | 41 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E51 | MEM A/BUS B2 | 42 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E52 | MEM A/BUS B2 | 42 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E53 | MEM A/BUS B2 | 42 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E54 | MEM A/BUS B3 | 43 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E55 | MEM A/BUS B3 | 43 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E56 | MEM A/BUS B3 | 43 | | | | | |
| E57 | IREFRESH REQ | 46 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E58 | MEM A/BUS B4 | 47 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E59 | MEM A/BUS B4 | 47 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E60 | MEM A/BUS B4 | 47 | 0.0 | -150.0 | -60.0 | mA | 1-4 |



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TABLE VI. Output short circuit current test - Continued.

| Test cond | Signal name | Pin no. | Vout (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| E61 | MEM A/BUS B5 | 48 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E62 | MEM A/BUS B5 | 48 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E63 | MEM A/BUS B5 | 48 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E64 | MEM A/BUS B6 | 49 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E65 | MEM A/BUS B6 | 49 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E66 | MEM A/BUS B6 | 49 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E67 | MEM A/BUS B7 | 50 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E68 | MEM A/BUS B7 | 50 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E69 | MEM A/BUS B7 | 50 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E70 | MEM A/BUS A8 | 52 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E71 | MEM A/BUS A8 | 52 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E72 | MEM A/BUS A8 | 52 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E73 | MEM A/BUS A9 | 53 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E74 | MEM A/BUS A9 | 53 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E75 | MEM A/BUS A9 | 53 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E76 | MEM A/BUSA10 | 54 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E77 | MEM A/BUSA10 | 54 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E78 | MEM A/BUSA10 | 54 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E79 | MEM A/BUSA11 | 55 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E80 | MEM A/BUSA11 | 55 | 0.0 | -150.0 | -60.0 | mA | 1-4 |



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TABLE VI. Output short circuit current test - Continued.

| Test cond | Signal name | Pin no. | Vout (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| E81 | MEM A/BUSA11 | 55 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E82 | MEM D/BUS 6 | 56 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E83 | MEM D/BUS 4 | 57 | | | | | |
| E84 | MEM D/BUS 2 | 58 | | | | | |
| E85 | MEM D/BUS 0 | 59 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E86 | MEM A/BUSA12 | 61 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E87 | MEM A/BUSA12 | 61 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E88 | MEM A/BUSA12 | 61 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E89 | MEM A/BUSA13 | 62 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E90 | MEM A/BUSA13 | 62 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E91 | MEM A/BUSA13 | 62 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E92 | MEM A/BUSA14 | 63 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E93 | MEM A/BUSA14 | 63 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E94 | MEM A/BUSA14 | 63 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E95 | MEM A/BUSA15 | 64 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E96 | MEM A/BUSA15 | 64 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E97 | MEM A/BUSA15 | 64 | | | | | |
| E98 | MEM A/BUSA18 | 65 | | | | | |
| E99 | MEM A/BUSA18 | 65 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E100 | MEM A/BUSA18 | 65 | 0.0 | -150.0 | -60.0 | mA | 1-4 |



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TABLE VI. Output short circuit current test - Continued.

| Test cond | Signal name | Pin no. | Vout (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| E101 | MEM A/BUSA19 | 66 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E102 | MEM A/BUSA19 | 66 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E103 | MEM A/BUSA19 | 66 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E104 | MEM D/BUS 8 | 67 | | | | | |
| E105 | MEM D/BUS 10 | 68 | | | | | |
| E106 | MEM D/BUS 12 | 69 | | | | | |
| E107 | MEM D/BUS 14 | 70 | | | | | |
| E108 | IN SCRUB REQ | 86 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E109 | MEM A/BUSB18 | 87 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E110 | MEM A/BUSB18 | 87 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E111 | MEM A/BUSB18 | 87 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E112 | MEM A/BUSB19 | 88 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E113 | MEM A/BUSB19 | 88 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E114 | MEM A/BUSB19 | 88 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E115 | MEM A/BUS B8 | 89 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E116 | MEM A/BUS B8 | 89 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E117 | MEM A/BUS B8 | 89 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E118 | MEM A/BUS B9 | 91 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E119 | MEM A/BUS B9 | 91 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E120 | MEM A/BUS B9 | 91 | 0.0 | -325.0 | -100.0 | mA | 1-4 |



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TABLE VI. Output short circuit current test - Continued.

| Test cond | Signal name | Pin no. | Vout (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| E121 | MEM A/BUSB10 | 92 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E122 | MEM A/BUSB10 | 92 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E123 | MEM A/BUSB10 | 92 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E124 | MEM A/BUSB11 | 93 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E125 | MEM A/BUSB11 | 93 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E126 | MEM A/BUSB11 | 93 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E127 | SCRUB CTR CA | 95 | 0.0 | -100.0 | -20.0 | mA | 1-4 |
| E128 | MEM A/BUSB12 | 97 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E129 | MEM A/BUSB12 | 97 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E130 | MEM A/BUSB12 | 97 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E131 | MEM A/BUSB13 | 98 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E132 | MEM A/BUSB13 | 98 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E133 | MEM A/BUSB13 | 98 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E134 | MEM A/BUSB14 | 99 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E135 | MEM A/BUSB14 | 99 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E136 | MEM A/BUSB14 | 99 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E137 | MEM A/BUSB15 | 100 | 0.0 | -325.0 | -100.0 | mA | 1-4 |
| E138 | MEM A/BUSB15 | 100 | 0.0 | -150.0 | -60.0 | mA | 1-4 |
| E139 | MEM A/BUSB15 | 100 | 0.0 | -325.0 | -100.0 | mA | 1-4 |



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NOTES:

1. Output short circuit current test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 5.5 V | ± 55 mV | E1-E139 |
| Power supply current limit | 2.45 A | NA | E1-E139 |
| Forcing voltage, (Vout) | 0.0 V | ± 50 mV | E1-E139 |
| Pattern voltage<br>high, (VH)<br>low, (VL) | <br>3.0 V<br>0.0 V | <br>± 100 mV<br>± 100 mV | <br>E1-E139<br>E1-E139 |
| Output current measurement | NA | ± 11.2 mA | E1 |
| | NA | ± 4.5 mA | E2,E3 |
| | NA | ± 11.2 mA | E4 |
| | NA | ± 4.5 mA | E5,E6 |
| | NA | ± 11.2 mA | E7 |
| | NA | ± 4.5 mA | E8,E9 |
| | NA | ± 11.2 mA | E10 |
| | NA | ± 4.5 mA | E11-E16 |
| | NA | ± 11.2 mA | E17 |
| | NA | ± 4.5 mA | E18,E19 |
| | NA | ± 11.2 mA | E20 |
| | NA | ± 4.5 mA | E21,E22 |
| | NA | ± 11.2 mA | E23 |
| | NA | ± 4.5 mA | E24,E25 |
| | NA | ± 11.2 mA | E26 |
| | NA | ± 4.5 mA | E27-E29 |
| | NA | ± 11.2 mA | E30,E31 |
| | NA | ± 4.5 mA | E32 |
| | NA | ± 11.2 mA | E33,E34 |
| | NA | ± 4.5 mA | E35-E39 |
| | NA | ± 11.2 mA | E40,E41 |
| | NA | ± 4.5 mA | E42 |
| | NA | ± 11.2 mA | E43-E45 |



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NOTES: - Continued.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Output current measurement | NA | ± 4.5 mA | E46,E47 |
| | NA | ± 11.2 mA | E48 |
| | NA | ± 4.5 mA | E49,E50 |
| | NA | ± 11.2 mA | E51 |
| | NA | ± 4.5 mA | E52,E53 |
| | NA | ± 11.2 mA | E54 |
| | NA | ± 4.5 mA | E55-E57 |
| | NA | ± 11.2 mA | E58 |
| | NA | ± 4.5 mA | E59,E60 |
| | NA | ± 11.2 mA | E61 |
| | NA | ± 4.5 mA | E62,E63 |
| | NA | ± 11.2 mA | E64 |
| | NA | ± 4.5 mA | E65,E66 |
| | NA | ± 11.2 mA | E67 |
| | NA | ± 4.5 mA | E68,E69 |
| | NA | ± 11.2 mA | E70 |
| | NA | ± 4.5 mA | E71 |
| | NA | ± 11.2 mA | E72,E73 |
| | NA | ± 4.5 mA | E74 |
| | NA | ± 11.2 mA | E75,E76 |
| | NA | ± 4.5 mA | E77 |
| | NA | ± 11.2 mA | E78,E79 |
| | NA | ± 4.5 mA | E80 |
| | NA | ± 11.2 mA | E81 |
| | NA | ± 4.5 mA | E82-E85 |
| | NA | ± 11.2 mA | E86 |
| | NA | ± 4.5 mA | E87 |
| | NA | ± 11.2 mA | E88,E89 |
| | NA | ± 4.5 mA | E90 |
| | NA | ± 11.2 mA | E91,E92 |
| | NA | ± 4.5 mA | E93 |
| | NA | ± 11.2 mA | E94 |
| | NA | ± 4.5 mA | E95 |
| | NA | ± 11.2 mA | E96-E99 |
| | NA | ± 4.5 mA | E100 |
| | NA | ± 11.2 mA | E101,E102 |
| | NA | ± 4.5 mA | E103-E108 |
| | NA | ± 11.2 mA | E109,E110 |
| | NA | ± 4.5 mA | E111 |
| | NA | ± 11.2 mA | E112,E113 |
| | NA | ± 4.5 mA , | E114 |
| | NA | ± 11.2 mA | E115 |



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NOTES: - Continued.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Output current measurement | NA | ± 4.5 mA | E116 |
| | NA | ± 11.2 mA | E117,E118 |
| | NA | ± 4.5 mA | E119 |
| | NA | ± 11.2 mA | E120,E121 |
| | NA | ± 4.5 mA | E122 |
| | NA | ± 11.2 mA | E123,E124 |
| | NA | ± 4.5 mA | E125 |
| | NA | ± 11.2 mA | E126 |
| | NA | ± 4.0 mA | E127 |
| | NA | ± 11.2 mA | E128 |
| | NA | ± 4.5 mA | E129 |
| | NA | ± 11.2 mA | E130,E131 |
| | NA | ± 4.5 mA | E132 |
| | NA | ± 11.2 mA | E133,E134 |
| | NA | ± 4.5 mA | E135 |
| | NA | ± 11.2 mA | E136,E137 |
| | NA | ± 4.5 mA | E138 |
| | NA | ± 11.2 mA | E139 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.



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NOTES: -Continued.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.



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TABLE VII. Input current, high level, test.

| Test cond | Signal name | Pin no. | Vih (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| F1 | MEM A/BUS A0 | 2 | 2.4 | -1.35 | 0.19 | mA | 1-3 |
| F2 | MEM A/BUS A1 | 3 | | | | | |
| F3 | MEM A/BUS A2 | 4 | | | | | |
| F4 | MEM A/BUS A3 | 5 | | | | | |
| F5 | MEM A/BUS A4 | 11 | | | | | |
| F6 | MEM A/BUS A5 | 12 | | | | | |
| F7 | MEM A/BUS A6 | 13 | | | | | |
| F8 | MEM A/BUS A7 | 14 | 2.4 | -1.35 | 0.19 | mA | 1-3 |
| F9 | MEM A/BUSA16 | 15 | 2.4 | -0.75 | 0.17 | mA | 1-3 |
| F10 | MEM A/BUSA17 | 16 | 2.4 | -0.75 | 0.17 | mA | 1-3 |
| F11 | LATCH ADRES | 21 | 2.4 | -4.8 | 0.16 | mA | 1-3 |
| F12 | LPBACK SEL 2 | 22 | 2.4 | -1.8 | 0.06 | mA | 1-3 |
| F13 | CB AD/WR D0 | 23 | 2.4 | -1.2 | 0.04 | mA | 1-3 |
| F14 | CB AD/WR D1 | 24 | | | | | |
| F15 | CB AD/WR D2 | 26 | | | | | |
| F16 | CB AD/WR D3 | 27 | | | | | |
| F17 | CB AD/WR D4 | 28 | | | | | |
| F18 | CB AD/WR D5 | 29 | | | | | |
| F19 | CB AD/WR D6 | 30 | | | | | |
| F20 | CB AD/WR D7 | 32 | 2.4 | -1.2 | 0.04 | mA | 1-3 |



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TABLE VII. Input current, high level, test - Continued.

| Test cond | Signal name | Pin no. | Vih (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| F21 | CB AD/WR D16 | 33 | 2.4 | -1.2 | 0.04 | mA | 1-3 |
| F22 | CB AD/WR D17 | 34 | | | | | |
| F23 | CB AD/WR D18 | 35 | | | | | |
| F24 | CB AD/WR D19 | 36 | 2.4 | -1.2 | 0.04 | mA | 1-3 |
| F25 | MEM A/BUSB16 | 37 | 2.4 | -0.75 | 0.17 | mA | 1-3 |
| F26 | MEM A/BUSB17 | 38 | 2.4 | -0.75 | 0.17 | mA | 1-3 |
| F27 | MEM A/BUS B0 | 39 | 2.4 | -1.35 | 0.19 | mA | 1-3 |
| F28 | MEM A/BUS B1 | 41 | | | | | |
| F29 | MEM A/BUS B2 | 42 | | | | | |
| F30 | MEM A/BUS B3 | 43 | 2.4 | -1.35 | 0.19 | mA | 1-3 |
| F31 | PWR UP MS CL | 44 | 2.4 | -0.6 | 0.02 | mA | 1-3 |
| F32 | GREFRESH REQ | 45 | 2.4 | -1.2 | 0.04 | mA | 1-3 |
| F33 | MEM A/BUS B4 | 47 | 2.4 | -1.35 | 0.19 | mA | 1-3 |
| F34 | MEM A/BUS B5 | 48 | | | | | |
| F35 | MEM A/BUS B6 | 49 | | | | | |
| F36 | MEM A/BUS B7 | 50 | 2.4 | -1.35 | 0.19 | mA | 1-3 |
| F37 | MEM A/BUS A8 | 52 | 2.4 | -0.75 | 0.17 | mA | 1-3 |
| F38 | MEM A/BUS A9 | 53 | | | | | |
| F39 | MEM A/BUSA10 | 54 | | | | | |
| F40 | MEM A/BUSA11 | 55 | 2.4 | -0.75 | 0.17 | mA | 1-3 |



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TABLE VII. Input current, high level, test - Continued.

| Test cond | Signal name | Pin no. | Vih (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| F41 | MEM A/BUSA12 | 61 | 2.4 | -0.75 | 0.17 | mA | 1-3 |
| F42 | MEM A/BUSA13 | 62 | | | | | |
| F43 | MEM A/BUSA14 | 63 | | | | | |
| F44 | MEM A/BUSA15 | 64 | | | | | |
| F45 | MEM A/BUSA18 | 65 | | | | | |
| F46 | MEM A/BUSA19 | 66 | 2.4 | -0.75 | 0.17 | mA | 1-3 |
| F47 | LPBACK SEL 1 | 71 | 2.4 | -4.2 | 0.14 | mA | 1-3 |
| F48 | LPBACK SEL 0 | 72 | 2.4 | -3.0 | 0.1 | mA | 1-3 |
| F49 | CB AD/WR D15 | 73 | 2.4 | -1.2 | 0.04 | mA | 1-3 |
| F50 | CB AD/WR D14 | 74 | | | | | |
| F51 | CB AD/WR D13 | 76 | | | | | |
| F52 | CB AD/WR D12 | 77 | | | | | |
| F53 | CB AD/WR D11 | 78 | | | | | |
| F54 | CB AD/WR D10 | 79 | | | | | |
| F55 | CB AD/WR D9 | 80 | | | | | |
| F56 | CB AD/WR D8 | 82 | 2.4 | -1.2 | 0.04 | mA | 1-3 |
| F57 | COM BUS CLK | 83 | 2.4 | -0.6 | 0.02 | mA | 1-3 |
| F58 | CL SCRUB CTR | 84 | 2.4 | -0.6 | 0.02 | mA | 1-3 |
| F59 | TEST CLOCK | 85 | 2.4 | -3.31 | -1.41 | mA | 1-3 |
| F60 | MEM A/BUSB18 | 87 | 2.4 | -0.75 | 0.17 | mA | 1-3 |



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TABLE VII. Input current, high level, test - Continued.

| Test cond | Signal name | Pin no. | Vih (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| F61 | MEM A/BUSB19 | 88 | 2.4 | -0.75 | 0.17 | mA | 1-3 |
| F62 | MEM A/BUS B8 | 89 | | | | | |
| F63 | MEM A/BUS B9 | 91 | | | | | |
| F64 | MEM A/BUSB10 | 92 | | | | | |
| F65 | MEM A/BUSB11 | 93 | 2.4 | -0.75 | 0.17 | mA | 1-3 |
| F66 | ADVSCRUB CTR | 94 | 2.4 | -0.6 | 0.02 | mA | 1-3 |
| F67 | FIRST WORD | 96 | 2.4 | -1.8 | 0.06 | mA | 1-3 |
| F68 | MEM A/BUSB12 | 97 | 2.4 | -0.75 | 0.17 | mA | 1-3 |
| F69 | MEM A/BUSB13 | 98 | | | | | |
| F70 | MEM A/BUSB14 | 99 | | | | | |
| F71 | MEM A/BUSB15 | 100 | 2.4 | -0.75 | 0.17 | mA | 1-3 |



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NOTES:

1. Input current, high level, test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 5.5 V | ± 55 mV | F1-F71 |
| Power supply current limit | 2.45 A | NA | F1-F71 |
| Forcing voltage, (Vih) | 2.4 V | ± 50 mV | F1-F71 |
| Pattern voltage | | | |
| high, (VH) | 3.0 V | ± 100 mV | F1-F71 |
| low, (VL) | 0.0 V | ± 100 mV | F1-F71 |
| Input current measurement | NA | ± 77 uA | F1-F8 |
| | NA | ± 46 uA | F9,F10 |
| | NA | ± 248 uA | F11 |
| | NA | ± 93 uA | F12 |
| | NA | ± 62 uA | F13-F24 |
| | NA | ± 46 uA | F25,F26 |
| | NA | ± 77 uA | F27-F30 |
| | NA | ± 31 uA | F31 |
| | NA | ± 62 uA | F32 |
| | NA | ± 77 uA | F33-F36 |
| | NA | ± 46 uA | F37-F46 |
| | NA | ± 217 uA | F47 |
| | NA | ± 155 uA | F48 |
| | NA | ± 62 uA | F49-F56 |
| | NA | ± 61 uA | F57,F58 |
| | NA | ± 95 uA | F59 |
| | NA | ± 46 uA | F60-F65 |
| | NA | ± 31 uA | F66 |
| | NA | ± 93 uA | F67 |
| | NA | ± 46 uA | F68-F71 |



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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.



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TABLE VIII. Input current, low level, test.

| Test cond | Signal name | Pin no. | Vil (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| G1 | MEM A/BUS A0 | 2 | 0.4 | -1.35 | 0.19 | mA | 1-3 |
| G2 | MEM A/BUS A1 | 3 | | | | | |
| G3 | MEM A/BUS A2 | 4 | | | | | |
| G4 | MEM A/BUS A3 | 5 | | | | | |
| G5 | MEM A/BUS A4 | 11 | | | | | |
| G6 | MEM A/BUS A5 | 12 | | | | | |
| G7 | MEM A/BUS A6 | 13 | | | | | |
| G8 | MEM A/BUS A7 | 14 | 0.4 | -1.35 | 0.19 | mA | 1-3 |
| G9 | MEM A/BUSA16 | 15 | 0.4 | -0.75 | 0.17 | mA | 1-3 |
| G10 | MEM A/BUSA17 | 16 | 0.4 | -0.75 | 0.17 | mA | 1-3 |
| G11 | LATCH ADRES | 21 | 0.4 | -4.8 | 0.16 | mA | 1-3 |
| G12 | LPBACK SEL 2 | 22 | 0.4 | -1.8 | 0.06 | mA | 1-3 |
| G13 | CB AD/WR D0 | 23 | 0.4 | -1.2 | 0.04 | mA | 1-3 |
| G14 | CB AD/WR D1 | 24 | | | | | |
| G15 | CB AD/WR D2 | 26 | | | | | |
| G16 | CB AD/WR D3 | 27 | | | | | |
| G17 | CB AD/WR D4 | 28 | | | | | |
| G18 | CB AD/WR D5 | 29 | | | | | |
| G19 | CB AD/WR D6 | 30 | | | | | |
| G20 | CB AD/WR D7 | 32 | 0.4 | -1.2 | 0.04 | mA | 1-3 |



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TABLE VIII. Input current, low level, test - Continued.

| Test cond | Signal name | Pin no. | Vil (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| G21 | CB AD/WR D16 | 33 | 0.4 | -1.2 | 0.04 | mA | 1-3 |
| G22 | CB AD/WR D17 | 34 | | | | | |
| G23 | CB AD/WR D18 | 35 | | | | | |
| G24 | CB AD/WR D19 | 36 | 0.4 | -1.2 | 0.04 | mA | 1-3 |
| G25 | MEM A/BUSB16 | 37 | 0.4 | -0.75 | 0.17 | mA | 1-3 |
| G26 | MEM A/BUSB17 | 38 | 0.4 | -0.75 | 0.17 | mA | 1-3 |
| G27 | MEM A/BUS B0 | 39 | 0.4 | -1.35 | 0.19 | mA | 1-3 |
| G28 | MEM A/BUS B1 | 41 | | | | | |
| G29 | MEM A/BUS B2 | 42 | | | | | |
| G30 | MEM A/BUS B3 | 43 | 0.4 | -1.35 | 0.19 | mA | 1-3 |
| G31 | PWR UP MS CL | 44 | 0.4 | -0.6 | 0.02 | mA | 1-3 |
| G32 | GREFRESH REQ | 45 | 0.4 | -1.2 | 0.04 | mA | 1-3 |
| G33 | MEM A/BUS B4 | 47 | 0.4 | -1.35 | 0.19 | mA | 1-3 |
| G34 | MEM A/BUS B5 | 48 | | | | | |
| G35 | MEM A/BUS B6 | 49 | | | | | |
| G36 | MEM A/BUS B7 | 50 | 0.4 | -1.35 | 0.19 | mA | 1-3 |
| G37 | MEM A/BUS A8 | 52 | 0.4 | -0.75 | 0.17 | mA | 1-3 |
| G38 | MEM A/BUS A9 | 53 | | | | | |
| G39 | MEM A/BUSA10 | 54 | | | | | |
| G40 | MEM A/BUSA11 | 55 | 0.4 | -0.75 | 0.17 | mA | 1-3 |



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TABLE VIII. Input current, low level, test - Continued.

| Test cond | Signal name | Pin no. | Vil (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| G41 | MEM A/BUSA12 | 61 | 0.4 | -0.75 | 0.17 | mA | 1-3 |
| G42 | MEM A/BUSA13 | 62 | | | | | |
| G43 | MEM A/BUSA14 | 63 | | | | | |
| G44 | MEM A/BUSA15 | 64 | | | | | |
| G45 | MEM A/BUSA18 | 65 | | | | | |
| G46 | MEM A/BUSA19 | 66 | 0.4 | -0.75 | 0.17 | mA | 1-3 |
| G47 | LPBACK SEL 1 | 71 | 0.4 | -4.2 | 0.14 | mA | 1-3 |
| G48 | LPBACK SEL 0 | 72 | 0.4 | -3.0 | 0.1 | mA | 1-3 |
| G49 | CB AD/WR D15 | 73 | 0.4 | -1.2 | 0.04 | mA | 1-3 |
| G50 | CB AD/WR D14 | 74 | | | | | |
| G51 | CB AD/WR D13 | 76 | | | | | |
| G52 | CB AD/WR D12 | 77 | | | | | |
| G53 | CB AD/WR D11 | 78 | | | | | |
| G54 | CB AD/WR D10 | 79 | | | | | |
| G55 | CB AD/WR D9 | 80 | | | | | |
| G56 | CB AD/WR D8 | 82 | 0.4 | -1.2 | 0.04 | mA | 1-3 |
| G57 | COM BUS CLK | 83 | 0.4 | -0.6 | 0.02 | mA | 1-3 |
| G58 | CL SCRUB CTR | 84 | 0.4 | -0.6 | 0.02 | mA | 1-3 |
| G59 | TEST CLOCK | 85 | 0.4 | -3.31 | -1.41 | mA | 1-3 |
| G60 | MEM A/BUSB18 | 87 | 0.4 | -0.75 | 0.17 | mA | 1-3 |



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TABLE VIII. Input current, low level, test - Continued.

| Test cond | Signal name | Pin no. | Vil (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| G61 | MEM A/BUSB19 | 88 | 0.4 | -0.75 | 0.17 | mA | 1-3 |
| G62 | MEM A/BUS B8 | 89 | | | | | |
| G63 | MEM A/BUS B9 | 91 | | | | | |
| G64 | MEM A/BUSB10 | 92 | | | | | |
| G65 | MEM A/BUSB11 | 93 | 0.4 | -0.75 | 0.17 | mA | 1-3 |
| G66 | ADVSCRUB CTR | 94 | 0.4 | -0.6 | 0.02 | mA | 1-3 |
| G67 | FIRST WORD | 96 | 0.4 | -1.8 | 0.06 | mA | 1-3 |
| G68 | MEM A/BUSB12 | 97 | 0.4 | -0.75 | 0.17 | mA | 1-3 |
| G69 | MEM A/BUSB13 | 98 | | | | | |
| G70 | MEM A/BUSB14 | 99 | | | | | |
| G71 | MEM A/BUSB15 | 100 | 0.4 | -0.75 | 0.17 | mA | 1-3 |



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NOTES:

1. Input current, low level, test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 5.5 V | ± 55 mV | G1-G71 |
| Power supply current limit | 2.45 A | NA | G1-G71 |
| Forcing voltage, (Vil) | 0.4 V | ± 50 mV | G1-G71 |
| Pattern voltage | | | |
| high, (VH) | 3.0 V | ± 100 mV | G1-G71 |
| low, (VL) | 0.0 V | ± 100 mV | G1-G71 |
| Input current measurement | NA | ± 77 uA | G1-G8 |
| | NA | ± 46 uA | G9,G10 |
| | NA | ± 248 uA | G11 |
| | NA | ± 93 uA | G12 |
| | NA | ± 62 uA | G13-G24 |
| | NA | ± 46 uA | G25,G26 |
| | NA | ± 77 uA | G27-G30 |
| | NA | ± 31 uA | G31 |
| | NA | ± 62 uA | G32 |
| | NA | ± 77 uA | G33-G36 |
| | NA | ± 46 uA | G37-G46 |
| | NA | ± 217 uA | G47 |
| | NA | ± 155 uA | G48 |
| | NA | ± 62 uA | G49-G56 |
| | NA | ± 31 uA | G57,G58 |
| | NA | ± 95 uA | G59 |
| | NA | ± 46 uA | G60-G65 |
| | NA | ± 31 uA | G66 |
| | NA | ± 93 uA | G67 |
| | NA | ± 46 uA | G68-G71 |



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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.



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TABLE IX. Output voltage, high level, test.

| Test cond | Signal name | Pin no. | Ioh (mA) | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| H1 | MEM A/BUS A0 | 2 | -0.4 | 2.4 | 4.5 | V | 1-4 |
| H2 | MEM A/BUS A0 | 2 | | | | | |
| H3 | MEM A/BUS A0 | 2 | | | | | |
| H4 | MEM A/BUS A1 | 3 | | | | | |
| H5 | MEM A/BUS A1 | 3 | | | | | |
| H6 | MEM A/BUS A1 | 3 | | | | | |
| H7 | MEM A/BUS A2 | 4 | | | | | |
| H8 | MEM A/BUS A2 | 4 | | | | | |
| H9 | MEM A/BUS A2 | 4 | | | | | |
| H10 | MEM A/BUS A3 | 5 | | | | | |
| H11 | MEM A/BUS A3 | 5 | | | | | |
| H12 | MEM A/BUS A3 | 5 | | | | | |
| H13 | MEM D/BUS 1 | 6 | | | | | |
| H14 | MEM D/BUS 3 | 7 | | | | | |
| H15 | MEM D/BUS 5 | 8 | | | | | |
| H16 | MEM D/BUS 7 | 9 | | | | | |
| H17 | MEM A/BUS A4 | 11 | | | | | |
| H18 | MEM A/BUS A4 | 11 | | | | | |
| H19 | MEM A/BUS A4 | 11 | | | | | |
| H20 | MEM A/BUS A5 | 12 | -0.4 | 2.4 | 4.5 | V | 1-4 |



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TABLE IX. Output voltage, high level, test - Continued.

| Test cond | Signal name | Pin no. | Ioh (mA) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| H21 | MEM A/BUS A5 | 12 | -0.4 | 2.4 | 4.5 | V | 1-4 |
| H22 | MEM A/BUS A5 | 12 | | | | | |
| H23 | MEM A/BUS A6 | 13 | | | | | |
| H24 | MEM A/BUS A6 | 13 | | | | | |
| H25 | MEM A/BUS A6 | 13 | | | | | |
| H26 | MEM A/BUS A7 | 14 | | | | | |
| H27 | MEM A/BUS A7 | 14 | | | | | |
| H28 | MEM A/BUS A7 | 14 | | | | | |
| H29 | MEM A/BUSA16 | 15 | | | | | |
| H30 | MEM A/BUSA16 | 15 | | | | | |
| H31 | MEM A/BUSA16 | 15 | | | | | |
| H32 | MEM A/BUSA17 | 16 | | | | | |
| H33 | MEM A/BUSA17 | 16 | | | | | |
| H34 | MEM A/BUSA17 | 16 | | | | | |
| H35 | MEM D/BUS 9 | 17 | | | | | |
| H36 | MEM D/BUS 11 | 18 | | | | | |
| H37 | MEM D/BUS 13 | 19 | | | | | |
| H38 | MEM D/BUS 15 | 20 | | | | | |
| H39 | MEM A/BUSB16 | 37 | | | | | |
| H40 | MEM A/BUSB16 | 37 | -0.4 | 2.4 | 4.5 | V | 1-4 |



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TABLE IX. Output voltage, high level, test - Continued.

| Test cond | Signal name | Pin no. | Ioh (mA) | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| H41 | MEM A/BUSB16 | 37 | -0.4 | 2.4 | 4.5 | V | 1-4 |
| H42 | MEM A/BUSB17 | 38 | | | | | |
| H43 | MEM A/BUSB17 | 38 | | | | | |
| H44 | MEM A/BUSB17 | 38 | | | | | |
| H45 | MEM A/BUS B0 | 39 | | | | | |
| H46 | MEM A/BUS B0 | 39 | | | | | |
| H47 | MEM A/BUS B0 | 39 | | | | | |
| H48 | MEM A/BUS B1 | 41 | | | | | |
| H49 | MEM A/BUS B1 | 41 | | | | | |
| H50 | MEM A/BUS B1 | 41 | | | | | |
| H51 | MEM A/BUS B2 | 42 | | | | | |
| H52 | MEM A/BUS B2 | 42 | | | | | |
| H53 | MEM A/BUS B2 | 42 | | | | | |
| H54 | MEM A/BUS B3 | 43 | | | | | |
| H55 | MEM A/BUS B3 | 43 | | | | | |
| H56 | MEM A/BUS B3 | 43 | | | | | |
| H57 | IREFRESH REQ | 46 | | | | | |
| H58 | MEM A/BUS B4 | 47 | | | | | |
| H59 | MEM A/BUS B4 | 47 | | | | | |
| H60 | MEM A/BUS B4 | 47 | -0.4 | 2.4 | 4.5 | V | 1-4 |



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TABLE IX. Output voltage, high level, test - Continued.

| Test cond | Signal name | Pin no. | Ioh (mA) | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| H61 | MEM A/BUS B5 | 48 | -0.4 | 2.4 | 4.5 | V | 1-4 |
| H62 | MEM A/BUS B5 | 48 | | | | | |
| H63 | MEM A/BUS B5 | 48 | | | | | |
| H64 | MEM A/BUS B6 | 49 | | | | | |
| H65 | MEM A/BUS B6 | 49 | | | | | |
| H66 | MEM A/BUS B6 | 49 | | | | | |
| H67 | MEM A/BUS B7 | 50 | | | | | |
| H68 | MEM A/BUS B7 | 50 | | | | | |
| H69 | MEM A/BUS B7 | 50 | | | | | |
| H70 | MEM A/BUS A8 | 52 | | | | | |
| H71 | MEM A/BUS A8 | 52 | | | | | |
| H72 | MEM A/BUS A8 | 52 | | | | | |
| H73 | MEM A/BUS A9 | 53 | | | | | |
| H74 | MEM A/BUS A9 | 53 | | | | | |
| H75 | MEM A/BUS A9 | 53 | | | | | |
| H76 | MEM A/BUSA10 | 54 | | | | | |
| H77 | MEM A/BUSA10 | 54 | | | | | |
| H78 | MEM A/BUSA10 | 54 | | | | | |
| H79 | MEM A/BUSA11 | 55 | | | | | |
| H80 | MEM A/BUSA11 | 55 | -0.4 | 2.4 | 4.5 | V | 1-4 |



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TABLE IX. Output voltage, high level, test - Continued.

| Test cond | Signal name | Pin no. | Ioh (mA) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| H81 | MEM A/BUSA11 | 55 | -0.4 | 2.4 | 4.5 | V | 1-4 |
| H82 | MEM D/BUS 6 | 56 | | | | | |
| H83 | MEM D/BUS 4 | 57 | | | | | |
| H84 | MEM D/BUS 2 | 58 | | | | | |
| H85 | MEM D/BUS 0 | 59 | | | | | |
| H86 | MEM A/BUSA12 | 61 | | | | | |
| H87 | MEM A/BUSA12 | 61 | | | | | |
| H88 | MEM A/BUSA12 | 61 | | | | | |
| H89 | MEM A/BUSA13 | 62 | | | | | |
| H90 | MEM A/BUSA13 | 62 | | | | | |
| H91 | MEM A/BUSA13 | 62 | | | | | |
| H92 | MEM A/BUSA14 | 63 | | | | | |
| H93 | MEM A/BUSA14 | 63 | | | | | |
| H94 | MEM A/BUSA14 | 63 | | | | | |
| H95 | MEM A/BUSA15 | 64 | | | | | |
| H96 | MEM A/BUSA15 | 64 | | | | | |
| H97 | MEM A/BUSA15 | 64 | | | | | |
| H98 | MEM A/BUSA18 | 65 | | | | | |
| H99 | MEM A/BUSA18 | 65 | | | | | |
| H100 | MEM A/BUSA18 | 65 | -0.4 | 2.4 | 4.5 | V | 1-4 |



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TABLE IX. Output voltage, high level, test - Continued.

| Test cond | Signal name | Pin no. | Ioh (mA) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| H101 | MEM A/BUSA19 | 66 | -0.4 | 2.4 | 4.5 | V | 1-4 |
| H102 | MEM A/BUSA19 | 66 | | | | | |
| H103 | MEM A/BUSA19 | 66 | | | | | |
| H104 | MEM D/BUS 8 | 67 | | | | | |
| H105 | MEM D/BUS 10 | 68 | | | | | |
| H106 | MEM D/BUS 12 | 69 | | | | | |
| H107 | MEM D/BUS 14 | 70 | | | | | |
| H108 | IN SCRUB REQ | 86 | | | | | |
| H109 | MEM A/BUSB18 | 87 | | | | | |
| H110 | MEM A/BUSB18 | 87 | | | | | |
| H111 | MEM A/BUSB18 | 87 | | | | | |
| H112 | MEM A/BUSB19 | 88 | | | | | |
| H113 | MEM A/BUSB19 | 88 | | | | | |
| H114 | MEM A/BUSB19 | 88 | | | | | |
| H115 | MEM A/BUS B8 | 89 | | | | | |
| H116 | MEM A/BUS B8 | 89 | | | | | |
| H117 | MEM A/BUS B8 | 89 | | | | | |
| H118 | MEM A/BUS B9 | 91 | | | | | |
| H119 | MEM A/BUS B9 | 91 | | | | | |
| H120 | MEM A/BUS B9 | 91 | -0.4 | 2.4 | 4.5 | V | 1-4 |



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TABLE IX. Output voltage, high level, test - Continued.

| Test cond | Signal name | Pin no. | Ioh (mA) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| H121 | MEM A/BUSB10 | 92 | -0.4 | 2.4 | 4.5 | V | 1-4 |
| H122 | MEM A/BUSB10 | 92 | | | | | |
| H123 | MEM A/BUSB10 | 92 | | | | | |
| H124 | MEM A/BUSB11 | 93 | | | | | |
| H125 | MEM A/BUSB11 | 93 | | | | | |
| H126 | MEM A/BUSB11 | 93 | -0.4 | 2.4 | 4.5 | V | 1-4 |
| H127 | SCRUB CTR CA | 95 | -0.16 | 2.4 | 4.5 | V | 1-4 |
| H128 | MEM A/BUSB12 | 97 | -0.4 | 2.4 | 4.5 | V | 1-4 |
| H129 | MEM A/BUSB12 | 97 | | | | | |
| H130 | MEM A/BUSB12 | 97 | | | | | |
| H131 | MEM A/BUSB13 | 98 | | | | | |
| H132 | MEM A/BUSB13 | 98 | | | | | |
| H133 | MEM A/BUSB13 | 98 | | | | | |
| H134 | MEM A/BUSB14 | 99 | | | | | |
| H135 | MEM A/BUSB14 | 99 | | | | | |
| H136 | MEM A/BUSB14 | 99 | | | | | |
| H137 | MEM A/BUSB15 | 100 | | | | | |
| H138 | MEM A/BUSB15 | 100 | | | | | |
| H139 | MEM A/BUSB15 | 100 | -0.4 | 2.4 | 4.5 | V | 1-4 |



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NOTES:

1. Output voltage, high level, test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 4.5 V | ± 45 mV | H1-H139 |
| Power supply current limit | 2.45 A | NA | H1-H139 |
| Forcing current, (Ioh) | -0.4 mA<br>-0.16 mA<br>-0.4 mA | ± 40 uA<br>± 16 uA<br>± 40 uA | H1-H126<br>H127<br>H128-H139 |
| Pattern voltage high, (VH)<br>low, (VL) | 3.0 V<br>0.0 V | ± 100 mV<br>± 100 mV | H1-H139<br>H1-H139 |
| Output voltage measurement | NA | ± 105 mV | H1-H139 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.



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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

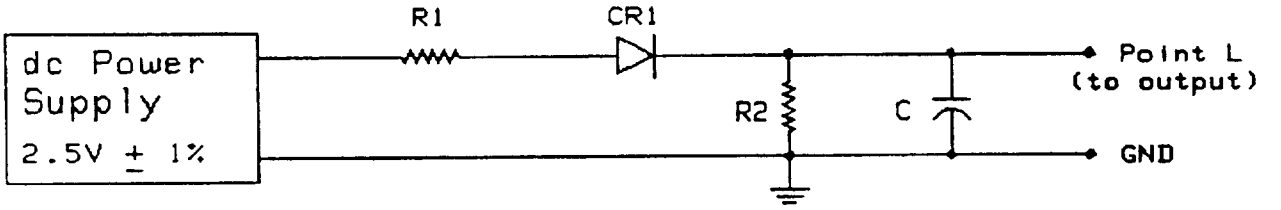
(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.



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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.



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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

| Pin number | R2 1/ (ohms) | Iol (mA) | Vol (volts) |
|---|---|---|---|
| 2,3,4,5,6,7,8,9, 11,12,13,14,15, 16,17,18,19,20, 37,38,39,41,42, 43,46,47,48,49, 50,52,53,54,55, 56,57,58,59,61, 62,63,64,65,66, 67,68,69,70,86, 87,88,89,91,92, 93,97,98,99,100 | 6200 | 6.0 | 0.5 |
| 95 | 16000 | 1.6 | 0.5 |

1/ All resistors are metal film, ± 1 percent, 1/4 W.



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TABLE X. Output voltage, low level, test.

| Test cond | Signal name | Pin no. | Iol (mA) | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| J1 | MEM A/BUS A0 | 2 | 6.0 | 0.0 | 0.5 | V | 1-4 |
| J2 | MEM A/BUS A0 | 2 | | | | | |
| J3 | MEM A/BUS A0 | 2 | | | | | |
| J4 | MEM A/BUS A1 | 3 | | | | | |
| J5 | MEM A/BUS A1 | 3 | | | | | |
| J6 | MEM A/BUS A1 | 3 | | | | | |
| J7 | MEM A/BUS A2 | 4 | | | | | |
| J8 | MEM A/BUS A2 | 4 | | | | | |
| J9 | MEM A/BUS A2 | 4 | | | | | |
| J10 | MEM A/BUS A3 | 5 | | | | | |
| J11 | MEM A/BUS A3 | 5 | | | | | |
| J12 | MEM A/BUS A3 | 5 | | | | | |
| J13 | MEM D/BUS 1 | 6 | | | | | |
| J14 | MEM D/BUS 3 | 7 | | | | | |
| J15 | MEM D/BUS 5 | 8 | | | | | |
| J16 | MEM D/BUS 7 | 9 | | | | | |
| J17 | MEM A/BUS A4 | 11 | | | | | |
| J18 | MEM A/BUS A4 | 11 | | | | | |
| J19 | MEM A/BUS A4 | 11 | | | | | |
| J20 | MEM A/BUS A5 | 12 | 6.0 | 0.0 | 0.5 | V | 1-4 |



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TABLE X. Output voltage, low level, test - Continued.

| Test cond | Signal name | Pin no. | Iol (mA) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| J21 | MEM A/BUS A5 | 12 | 6.0 | 0.0 | 0.5 | V | 1-4 |
| J22 | MEM A/BUS A5 | 12 | | | | | |
| J23 | MEM A/BUS A6 | 13 | | | | | |
| J24 | MEM A/BUS A6 | 13 | | | | | |
| J25 | MEM A/BUS A6 | 13 | | | | | |
| J26 | MEM A/BUS A7 | 14 | | | | | |
| J27 | MEM A/BUS A7 | 14 | | | | | |
| J28 | MEM A/BUS A7 | 14 | | | | | |
| J29 | MEM A/BUSA16 | 15 | | | | | |
| J30 | MEM A/BUSA16 | 15 | | | | | |
| J31 | MEM A/BUSA16 | 15 | | | | | |
| J32 | MEM A/BUSA17 | 16 | | | | | |
| J33 | MEM A/BUSA17 | 16 | | | | | |
| J34 | MEM A/BUSA17 | 16 | | | | | |
| J35 | MEM D/BUS 9 | 17 | | | | | |
| J36 | MEM D/BUS 11 | 18 | | | | | |
| J37 | MEM D/BUS 13 | 19 | | | | | |
| J38 | MEM D/BUS 15 | 20 | | | | | |
| J39 | MEM A/BUSB16 | 37 | | | | | |
| J40 | MEM A/BUSB16 | 37 | 6.0 | 0.0 | 0.5 | V | 1-4 |



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TABLE X. Output voltage, low level, test - Continued.

| Test cond | Signal name | Pin no. | Iol (mA) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| J41 | MEM A/BUSB16 | 37 | 6.0 | 0.0 | 0.5 | V | 1-4 |
| J42 | MEM A/BUSB17 | 38 | | | | | |
| J43 | MEM A/BUSB17 | 38 | | | | | |
| J44 | MEM A/BUSB17 | 38 | | | | | |
| J45 | MEM A/BUS B0 | 39 | | | | | |
| J46 | MEM A/BUS B0 | 39 | | | | | |
| J47 | MEM A/BUS B0 | 39 | | | | | |
| J48 | MEM A/BUS B1 | 41 | | | | | |
| J49 | MEM A/BUS B1 | 41 | | | | | |
| J50 | MEM A/BUS B1 | 41 | | | | | |
| J51 | MEM A/BUS B2 | 42 | | | | | |
| J52 | MEM A/BUS B2 | 42 | | | | | |
| J53 | MEM A/BUS B2 | 42 | | | | | |
| J54 | MEM A/BUS B3 | 43 | | | | | |
| J55 | MEM A/BUS B3 | 43 | | | | | |
| J56 | MEM A/BUS B3 | 43 | | | | | |
| J57 | IREFRESH REQ | 46 | | | | | |
| J58 | MEM A/BUS B4 | 47 | | | | | |
| J59 | MEM A/BUS B4 | 47 | | | | | |
| J60 | MEM A/BUS B4 | 47 | 6.0 | 0.0 | 0.5 | V | 1-4 |



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TABLE X. Output voltage, low level, test - Continued.

| Test cond | Signal name | Pin no. | Iol (mA) | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| J61 | MEM A/BUS B5 | 48 | 6.0 | 0.0 | 0.5 | V | 1-4 |
| J62 | MEM A/BUS B5 | 48 | | | | | |
| J63 | MEM A/BUS B5 | 48 | | | | | |
| J64 | MEM A/BUS B6 | 49 | | | | | |
| J65 | MEM A/BUS B6 | 49 | | | | | |
| J66 | MEM A/BUS B6 | 49 | | | | | |
| J67 | MEM A/BUS B7 | 50 | | | | | |
| J68 | MEM A/BUS B7 | 50 | | | | | |
| J69 | MEM A/BUS B7 | 50 | | | | | |
| J70 | MEM A/BUS A8 | 52 | | | | | |
| J71 | MEM A/BUS A8 | 52 | | | | | |
| J72 | MEM A/BUS A8 | 52 | | | | | |
| J73 | MEM A/BUS A9 | 53 | | | | | |
| J74 | MEM A/BUS A9 | 53 | | | | | |
| J75 | MEM A/BUS A9 | 53 | | | | | |
| J76 | MEM A/BUSA10 | 54 | | | | | |
| J77 | MEM A/BUSA10 | 54 | | | | | |
| J78 | MEM A/BUSA10 | 54 | | | | | |
| J79 | MEM A/BUSA11 | 55 | | | | | |
| J80 | MEM A/BUSA11 | 55 | 6.0 | 0.0 | 0.5 | V | 1-4 |



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TABLE X. Output voltage, low level, test - Continued.

| Test cond | Signal name | Pin no. | Iol (mA) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| J81 | MEM A/BUSA11 | 55 | 6.0 | 0.0 | 0.5 | V | 1-4 |
| J82 | MEM D/BUS 6 | 56 | | | | | |
| J83 | MEM D/BUS 4 | 57 | | | | | |
| J84 | MEM D/BUS 2 | 58 | | | | | |
| J85 | MEM D/BUS 0 | 59 | | | | | |
| J86 | MEM A/BUSA12 | 61 | | | | | |
| J87 | MEM A/BUSA12 | 61 | | | | | |
| J88 | MEM A/BUSA12 | 61 | | | | | |
| J89 | MEM A/BUSA13 | 62 | | | | | |
| J90 | MEM A/BUSA13 | 62 | | | | | |
| J91 | MEM A/BUSA13 | 62 | | | | | |
| J92 | MEM A/BUSA14 | 63 | | | | | |
| J93 | MEM A/BUSA14 | 63 | | | | | |
| J94 | MEM A/BUSA14 | 63 | | | | | |
| J95 | MEM A/BUSA15 | 64 | | | | | |
| J96 | MEM A/BUSA15 | 64 | | | | | |
| J97 | MEM A/BUSA15 | 64 | | | | | |
| J98 | MEM A/BUSA18 | 65 | | | | | |
| J99 | MEM A/BUSA18 | 65 | | | | | |
| J100 | MEM A/BUSA18 | 65 | 6.0 | 0.0 | 0.5 | V | 1-4 |



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TABLE X. Output voltage, low level, test - Continued.

| Test cond | Signal name | Pin no. | Iol (mA) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| J101 | MEM A/BUSA19 | 66 | 6.0 | 0.0 | 0.5 | V | 1-4 |
| J102 | MEM A/BUSA19 | 66 | | | | | |
| J103 | MEM A/BUSA19 | 66 | | | | | |
| J104 | MEM D/BUS 8 | 67 | | | | | |
| J105 | MEM D/BUS 10 | 68 | | | | | |
| J106 | MEM D/BUS 12 | 69 | | | | | |
| J107 | MEM D/BUS 14 | 70 | | | | | |
| J108 | IN SCRUB REQ | 86 | | | | | |
| J109 | MEM A/BUSB18 | 87 | | | | | |
| J110 | MEM A/BUSB18 | 87 | | | | | |
| J111 | MEM A/BUSB18 | 87 | | | | | |
| J112 | MEM A/BUSB19 | 88 | | | | | |
| J113 | MEM A/BUSB19 | 88 | | | | | |
| J114 | MEM A/BUSB19 | 88 | | | | | |
| J115 | MEM A/BUS B8 | 89 | | | | | |
| J116 | MEM A/BUS B8 | 89 | | | | | |
| J117 | MEM A/BUS B8 | 89 | | | | | |
| J118 | MEM A/BUS B9 | 91 | | | | | |
| J119 | MEM A/BUS B9 | 91 | | | | | |
| J120 | MEM A/BUS B9 | 91 | 6.0 | 0.0 | 0.5 | V | 1-4 |



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TABLE X. Output voltage, low level, test - Continued.

| Test cond | Signal name | Pin no. | Iol (mA) | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| J121 | MEM A/BUSB10 | 92 | 6.0 | 0.0 | 0.5 | V | 1-4 |
| J122 | MEM A/BUSB10 | 92 | | | | | |
| J123 | MEM A/BUSB10 | 92 | | | | | |
| J124 | MEM A/BUSB11 | 93 | | | | | |
| J125 | MEM A/BUSB11 | 93 | | | | | |
| J126 | MEM A/BUSB11 | 93 | 6.0 | 0.0 | 0.5 | V | 1-4 |
| J127 | SCRUB CTR CA | 95 | 1.6 | 0.0 | 0.5 | V | 1-4 |
| J128 | MEM A/BUSB12 | 97 | 6.0 | 0.0 | 0.5 | V | 1-4 |
| J129 | MEM A/BUSB12 | 97 | | | | | |
| J130 | MEM A/BUSB12 | 97 | | | | | |
| J131 | MEM A/BUSB13 | 98 | | | | | |
| J132 | MEM A/BUSB13 | 98 | | | | | |
| J133 | MEM A/BUSB13 | 98 | | | | | |
| J134 | MEM A/BUSB14 | 99 | | | | | |
| J135 | MEM A/BUSB14 | 99 | | | | | |
| J136 | MEM A/BUSB14 | 99 | | | | | |
| J137 | MEM A/BUSB15 | 100 | | | | | |
| J138 | MEM A/BUSB15 | 100 | | | | | |
| J139 | MEM A/BUSB15 | 100 | 6.0 | 0.0 | 0.5 | V | 1-4 |



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NOTES:

1. Output voltage, low level, test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 4.5 V | ± 45 mV | J1-J139 |
| Power supply current limit | 2.45 A | NA | J1-J139 |
| Forcing current, (Iol) | 6.0 mA<br>1.6 mA<br>6.0 mA | ± 600 uA<br>± 160 uA<br>± 600 uA | J1-J126<br>J127<br>J128-J139 |
| Pattern voltage<br>high, (VH)<br>low, (VL) | <br>3.0 V<br>0.0 V | <br>± 100 mV<br>± 100 mV | <br>J1-J139<br>J1-J139 |
| Output voltage measurement | NA | ± 25 mV | J1-J139 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.



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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.



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TABLE XI. Output leakage current, high and low level, test.

| Test cond | Signal name | Pin no. | Forcing voltage (V) | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| High Level | | | | | | | |
| K1 | MEM D/BUS 1 | 6 | 2.4 | -0.05 | 0.05 | mA | 1-3 |
| K2 | MEM D/BUS 3 | 7 | | | | | |
| K3 | MEM D/BUS 5 | 8 | | | | | |
| K4 | MEM D/BUS 7 | 9 | | | | | |
| K5 | MEM D/BUS 9 | 17 | | | | | |
| K6 | MEM D/BUS 11 | 18 | | | | | |
| K7 | MEM D/BUS 13 | 19 | | | | | |
| K8 | MEM D/BUS 15 | 20 | | | | | |
| K9 | MEM D/BUS 6 | 56 | | | | | |
| K10 | MEM D/BUS 4 | 57 | | | | | |
| K11 | MEM D/BUS 2 | 58 | | | | | |
| K12 | MEM D/BUS 0 | 59 | | | | | |
| K13 | MEM D/BUS 8 | 67 | | | | | |
| K14 | MEM D/BUS 10 | 68 | | | | | |
| K15 | MEM D/BUS 12 | 69 | | | | | |
| K16 | MEM D/BUS 14 | 70 | 2.4 | -0.05 | 0.05 | mA | 1-3 |
| Low Level | | | | | | | |
| K17 | MEM D/BUS 1 | 6 | 0.4 | -0.05 | 0.05 | mA | 1-3 |
| K18 | MEM D/BUS 3 | 7 | | | | | |
| K19 | MEM D/BUS 5 | 8 | 0.4 | -0.05 | 0.05 | mA | 1-3 |



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TABLE XI. Output leakage current, high and low level, test - Continued.

| Test cond | Signal name | Pin no. | Forcing voltage (V) | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| K20 | MEM D/BUS 7 | 9 | 0.4 | -0.05 | 0.05 | mA | 1-3 |
| K21 | MEM D/BUS 9 | 17 | | | | | |
| K22 | MEM D/BUS 11 | 18 | | | | | |
| K23 | MEM D/BUS 13 | 19 | | | | | |
| K24 | MEM D/BUS 15 | 20 | | | | | |
| K25 | MEM D/BUS 6 | 56 | | | | | |
| K26 | MEM D/BUS 4 | 57 | | | | | |
| K27 | MEM D/BUS 2 | 58 | | | | | |
| K28 | MEM D/BUS 0 | 59 | | | | | |
| K29 | MEM D/BUS 8 | 67 | | | | | |
| K30 | MEM D/BUS 10 | 68 | | | | | |
| K31 | MEM D/BUS 12 | 69 | | | | | |
| K32 | MEM D/BUS 14 | 70 | 0.4 | -0.05 | 0.05 | mA | 1-3 |



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NOTES:

1. Output leakage current, high and low level, test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 5.5 V | ± 55 mV | K1-K32 |
| Power supply current limit | 2.45 A | NA | K1-K32 |
| Forcing voltage,<br>(Voh)<br>(Vol) | <br>2.4 V<br>0.4 V | <br>± 50 mV<br>± 50 mV | <br>K1-K16<br>K17-K32 |
| Pattern voltage<br>high, (VH)<br>low, (VL) | <br>3.0 V<br>0.0 V | <br>± 100 mV<br>± 100 mV | <br>K1-K32<br>K1-K32 |
| Output current measurement | NA | ± 5 uA | K1-K32 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.



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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.



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TABLE XII. Input clamp diode voltage test.

| Test cond | Signal name | Pin no. | Iin (mA) | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| L1 | LATCH ADRES | 21 | -18.0 | -1.2 | 0.0 | V | 1-3 |
| L2 | LPBACK SEL 2 | 22 | | | | | |
| L3 | CB AD/WR D0 | 23 | | | | | |
| L4 | CB AD/WR D1 | 24 | | | | | |
| L5 | CB AD/WR D2 | 26 | | | | | |
| L6 | CB AD/WR D3 | 27 | | | | | |
| L7 | CB AD/WR D4 | 28 | | | | | |
| L8 | CB AD/WR D5 | 29 | | | | | |
| L9 | CB AD/WR D6 | 30 | | | | | |
| L10 | CB AD/WR D7 | 32 | | | | | |
| L11 | CB AD/WR D16 | 33 | | | | | |
| L12 | CB AD/WR D17 | 34 | | | | | |
| L13 | CB AD/WR D18 | 35 | | | | | |
| L14 | CB AD/WR D19 | 36 | | | | | |
| L15 | PWR UP MS CL | 44 | | | | | |
| L16 | GREFRESH REQ | 45 | | | | | |
| L17 | LPBACK SEL 1 | 71 | | | | | |
| L18 | LPBACK SEL 0 | 72 | | | | | |
| L19 | CB AD/WR D15 | 73 | | | | | |
| L20 | CB AD/WR D14 | 74 | -18.0 | -1.2 | 0.0 | V | 1-3 |



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TABLE XII. Input clamp diode voltage test - Continued.

| Test cond | Signal name | Pin no. | Iin (mA) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| L21 | CB AD/WR D13 | 76 | -18.0 | -1.2 | 0.0 | V | 1-3 |
| L22 | CB AD/WR D12 | 77 | | | | | |
| L23 | CB AD/WR D11 | 78 | | | | | |
| L24 | CB AD/WR D10 | 79 | | | | | |
| L25 | CB AD/WR D9 | 80 | | | | | |
| L26 | CB AD/WR D8 | 82 | | | | | |
| L27 | COM BUS CLK | 83 | | | | | |
| L28 | CL SCRUB CTR | 84 | | | | | |
| L29 | TEST CLOCK | 85 | | | | | |
| L30 | ADVSCRUB CTR | 94 | | | | | |
| L31 | FIRST WORD | 96 | -18.0 | -1.2 | 0.0 | V | 1-3 |



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NOTES:

1. Input clamp diode voltage test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | Off | NA | L1-L31 |
| Power supply current limit | NA | NA | L1-L31 |
| Forcing current, (Iin) | -18 mA | ± 1.8 mA | L1-L31 |
| Pattern voltage<br>high, (VH)<br>low, (VL) | <br>NA<br>NA | <br>NA<br>NA | <br>L1-L31<br>L1-L31 |
| Input voltage measurement | NA | ± 60 mV | L1-L31 |

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.



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TABLE XIII. Power supply filter capacitance test.

| Test cond | Parameter | Pin numbers | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|
| M1 | PSFC | 1 | 5.6 | 10.4 | uF | 1-4 |
| M2 | PSFC | 51 | 7.0 | 13.0 | uF | 1-4 |

NOTES:

1. Power supply filter capacitance test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Capacitance bridge frequency | 1 KHz | ± 100 Hz | M1,M2 |
| Capacitance bridge voltage | 0.3 Vp | ± 30 mVp | M1,M2 |
| Capacitance measurement | NA<br>NA | ± 240 nF<br>± 300 nF | M1<br>M2 |

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.



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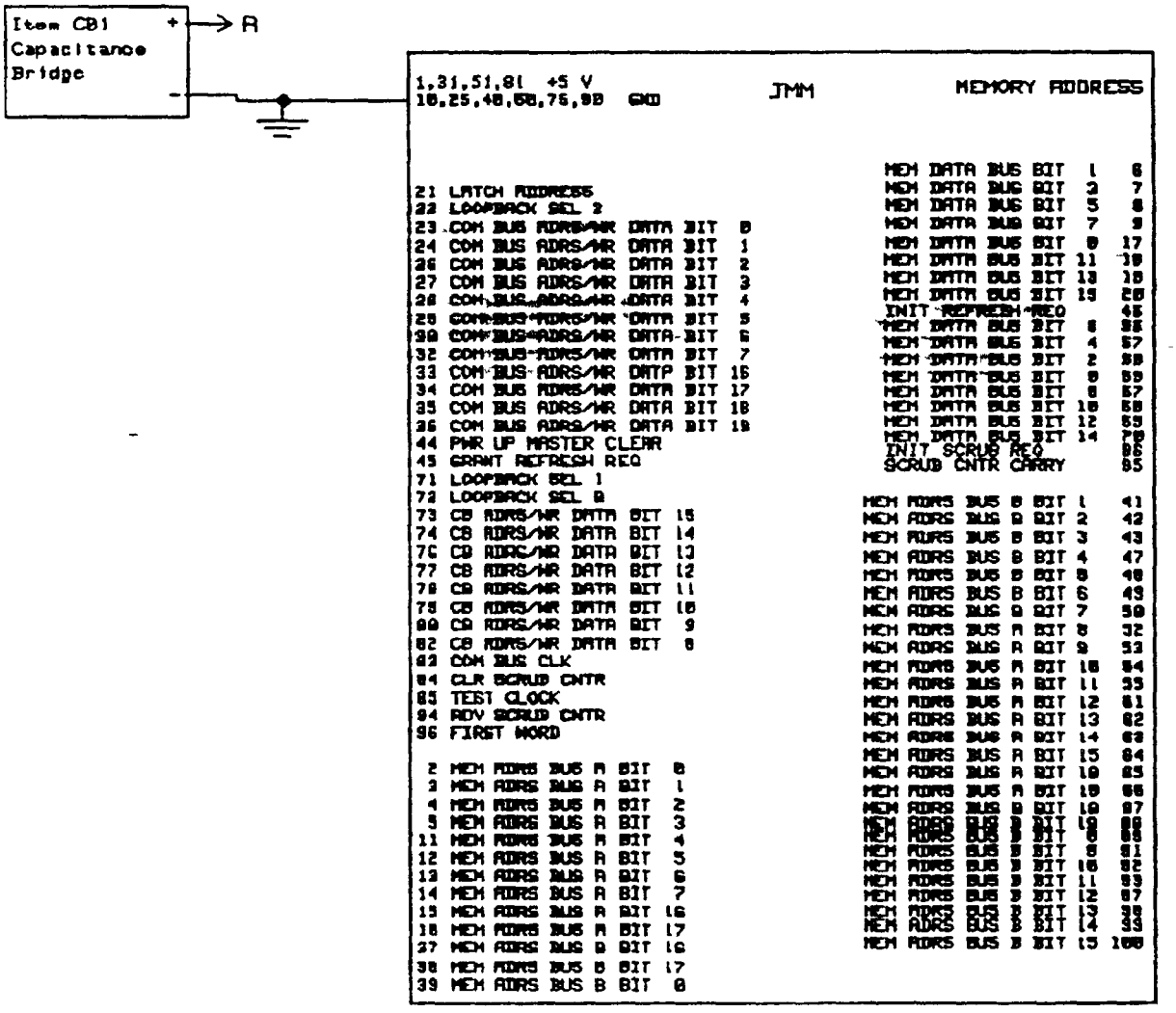



FIGURE 10. Power supply filter capacitance test circuit for table XIII.



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TABLE XIV. Power supply current test.

| Test cond | Parameter | Pin no. | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|
| N | Icc | 1,31, 51,81 | 680 | 1633 | mA | 1-3 |

NOTES:

1. Power supply current test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 5.5 V | ± 55 mV | N |
| Power supply current limit | 2.45 A | NA | N |
| Pattern voltage<br>high, (VH)<br>low, (VL) | <br>3.0 V<br>0.0 V | <br>± 100 mV<br>± 100 mV | <br>N<br>N |
| Power supply current measurement | NA | ± 47.6 mA | N |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.



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TABLE XV. Functional test.

| Test cond | Test description | Notes |
|---|---|---|
| P1 | Functional test<br>minimum power supply voltage<br>maximum pattern rate (425 nS) | 1-3 |
| P2 | Functional test<br>maximum power supply voltage<br>maximum pattern rate (425 nS) | 1-3 |

NOTES:

1. Functional test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 4.5 V<br>5.5 V | ± 45 mV<br>± 55 mV | P1<br>P2 |
| Power supply current limit | 2.45 A | NA | P1,P2 |
| Pattern voltage<br>high, (VH)<br>low, (VL) | <br>2.4 V<br>0.4 V | <br>± 100 mV<br>± 100 mV | <br>P1,P2<br>P1,P2 |
| Output voltage verification<br>high, (Voh)<br>high, (Voh)<br>low, (Vol) | <br>NA<br>NA<br>NA | <br>± 105 mV<br>± 155 mV<br>± 20 mV | <br>P1<br>P2<br>P1,P2 |



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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

3. Abbreviations used are defined in 6.1.



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TABLE XVI. Dynamic test.

| Test cond | Test description | Notes |
|---|---|---|
| Q1 | Dynamic test, sample<br>maximum pattern rate (425 nS)<br>minimum power supply | 1-6 |
| Q2 | Dynamic test, sample<br>maximum pattern rate (425 nS)<br>maximum power supply | 1-6 |
| Q3 | Dynamic test, qualification<br>maximum pattern rate (425 nS)<br>minimum power supply | 1-6 |
| Q4 | Dynamic test, qualification<br>maximum pattern rate (425 nS)<br>maximum power supply | 1-6 |

NOTES:

1. Dynamic test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 4.5 V<br>5.5 V | ± 45 mV<br>± 55 mV | Q1,Q3<br>Q2,Q4 |
| Power supply current limit | 2.45 A | NA | Q1-Q4 |
| Input voltage<br>high, (VH)<br>low, (VL) | <br>2.4 V<br>0.4 V | <br>± 100 mV<br>± 100 mV | <br>Q1-Q4<br>Q1-Q4 |



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NOTES: - Continued.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Output voltage verification | | | |
| high, (Voh) | NA | ± 100 mV | Q1,Q3 |
| high, (Voh) | NA | ± 100 mV | Q2,Q4 |
| low, (Vol) | NA | ± 100 mV | Q1-Q4 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

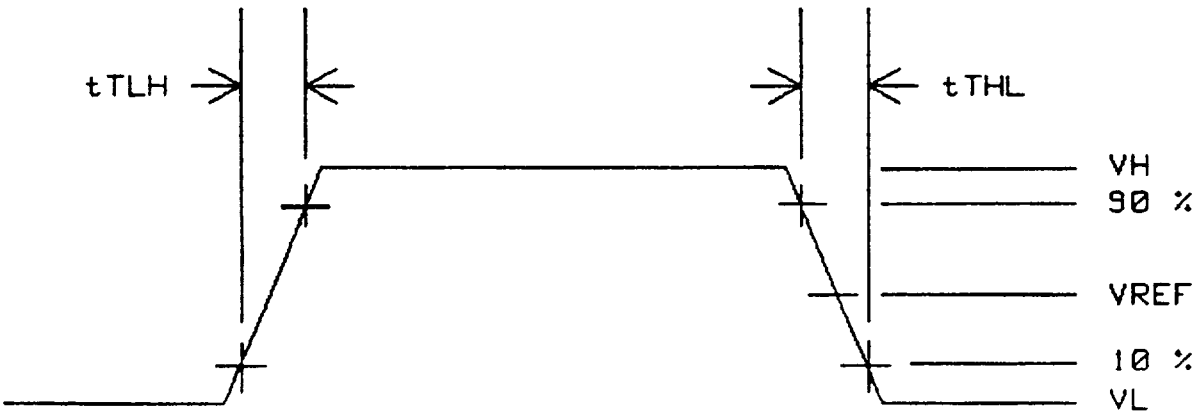
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 11. All of the input waveforms shall have the following characteristics:






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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.



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TABLE XVII. Dynamic parameters.

| Ref num | Sym | Parameter | Minimum limit | | | Maximum limit | | | Unit | Figs |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | -55°C | 25°C | 85°C | -55°C | 25°C | 85°C | | |
| 1 | tpd | LPBACK SEL 0,1 to MEM D/BUS 7-4 | - | - | - | 22 | 22 | 22 | nS | 11 |
| 2 | tpd | LPBACK SEL 2 to MEM D/BUS 15-0 | - | - | - | 32 | 32 | 32 | nS | 11 |
| 3 | tpd | GREFRESH REQ to IREFRESH REQ | - | - | - | 41 | 41 | 41 | nS | 11 |
| 4 | tpd | GREFRESH REQ to IN SCRUB REQ | - | - | - | 52 | 52 | 52 | nS | 11 |
| 5 | tpd | LPBACK SEL 1 to MEM A/BUS A19-0 | - | - | - | 51 | 51 | 51 | nS | 11 |
| 5 | tpd | LPBACK SEL 1 to MEM A/BUS B19-0 | - | - | - | 51 | 51 | 51 | nS | 11 |
| 6 | tpd | FIRST WORD to MEM A/BUS A15-0 | - | - | - | 36 | 36 | 36 | nS | 11 |
| 6 | tpd | FIRST WORD to MEM A/BUS B15-0 | - | - | - | 36 | 36 | 36 | nS | 11 |



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TABLE XVII. Dynamic parameters - Continued.

| Ref num | Sym | Parameter | Minimum limit | | | Maximum limit | | | Unit | Figs |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | -55°C | 25°C | 85°C | -55°C | 25°C | 85°C | | |
| 7 | tpd | LPBACK SEL 0 to MEM A/BUS A19-8 | - | - | - | 51 | 51 | 51 | nS | 11 |
| 7 | tpd | LPBACK SEL 0 to MEM A/BUS B19-8 | - | - | - | 51 | 51 | 51 | nS | 11 |
| 8 | tpd | LATCH ADRES to MEM A/BUS A19-0 | - | - | - | 25 | 25 | 25 | nS | 11 |
| 8 | tpd | LATCH ADRES to MEM A/BUS B19-0 | - | - | - | 25 | 25 | 25 | nS | 11 |
| 9 | tpd | CB AD/WR D19-0 to MEM A/BUS A19-0 | - | - | - | 22 | 22 | 22 | nS | 11 |
| 9 | tpd | CB AD/WR D19-0 to MEM A/BUS B19-0 | - | - | - | 22 | 22 | 22 | nS | 11 |



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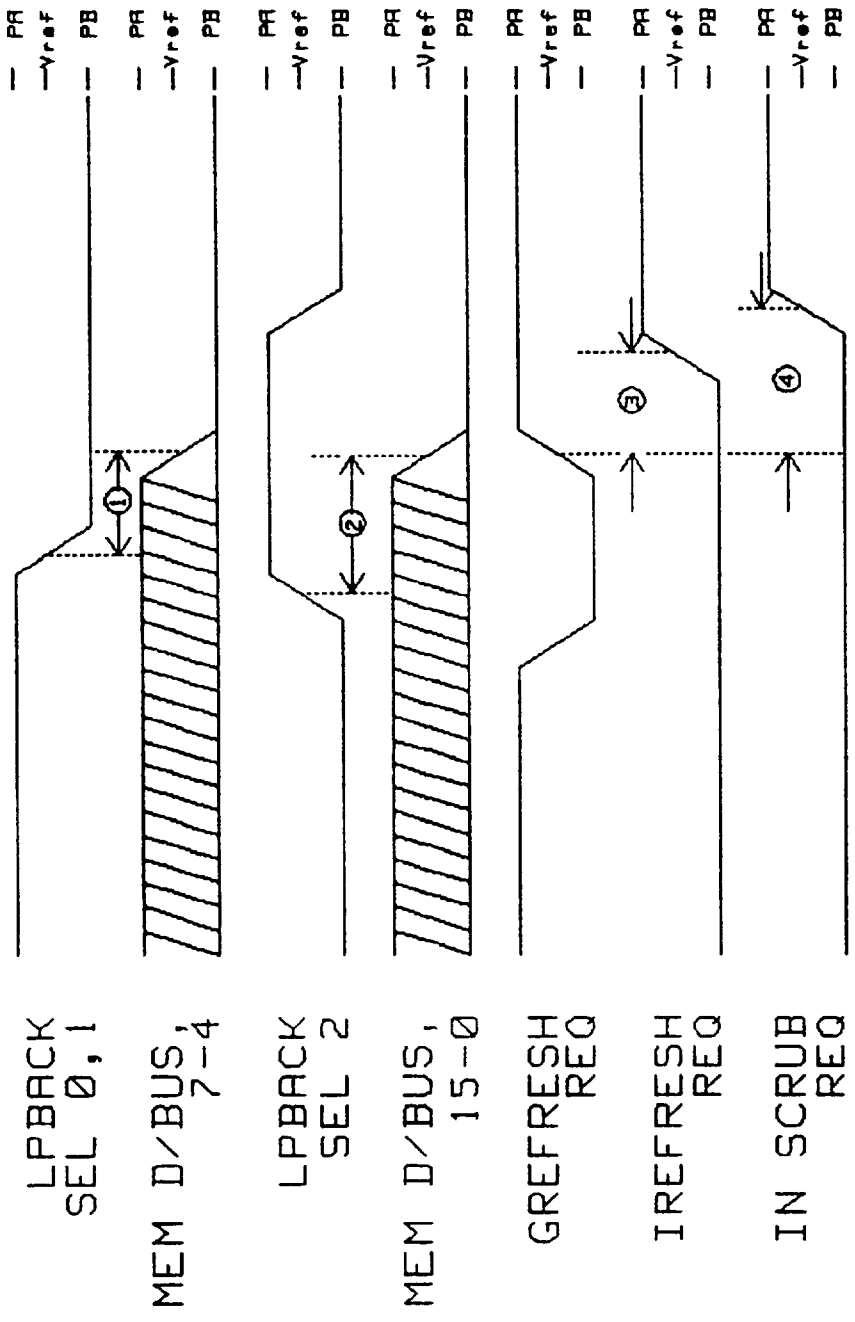



FIGURE 11a. Dynamic waveforms for table XVII.



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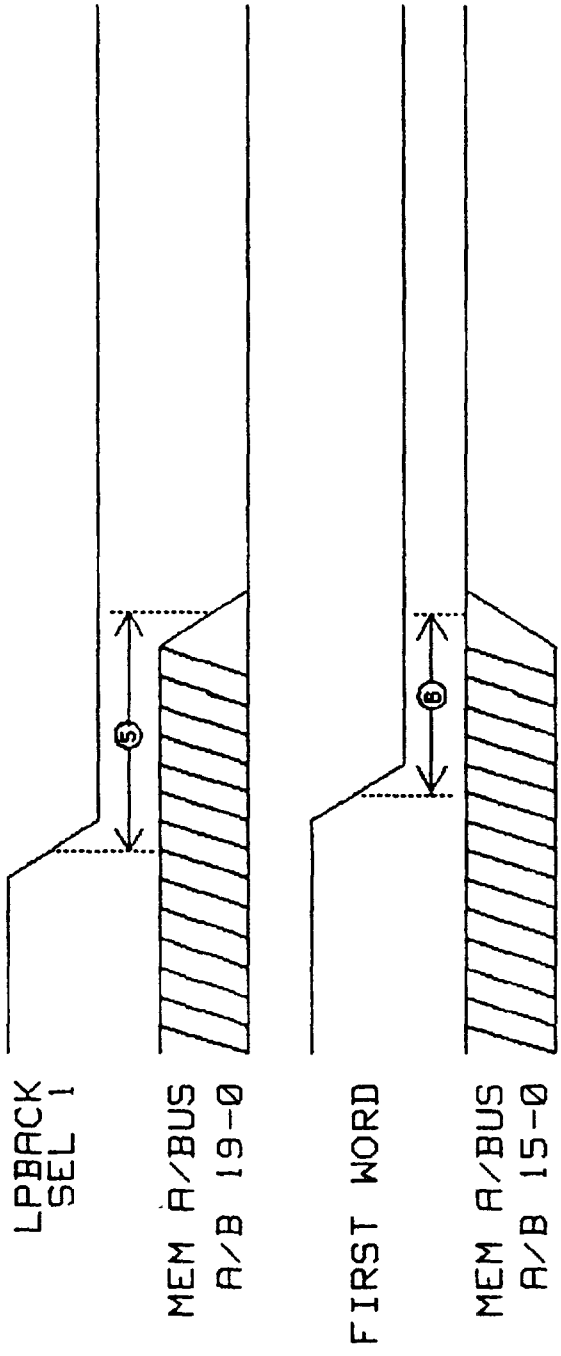



FIGURE 11b. Dynamic waveforms for table XVII.



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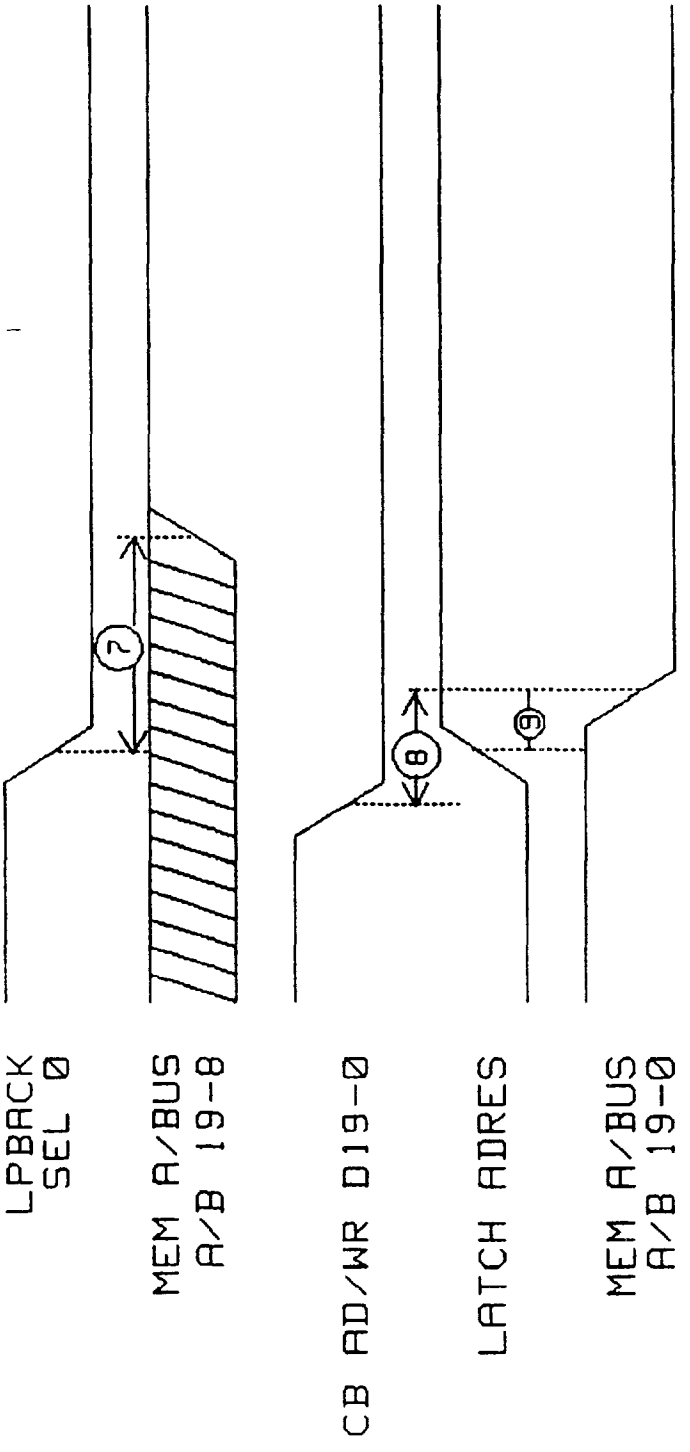



FIGURE 11c. Dynamic waveforms for table XVII.



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TABLE XVIII. Input and output capacitance test.

| Test cond | Signal name | Pin no. | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|
| R1 | LATCH ADRES | 21 | 12 | 216 | pF | 1-4 |
| R2 | LPBACK SEL 2 | 22 | 6 | 100 | pF | 1-4 |
| R3 | CB AD/WR D0 | 23 | 4 | 100 | pF | 1-4 |
| R4 | CB AD/WR D1 | 24 | | | | |
| R5 | CB AD/WR D2 | 26 | | | | |
| R6 | CB AD/WR D3 | 27 | | | | |
| R7 | CB AD/WR D4 | 28 | | | | |
| R8 | CB AD/WR D5 | 29 | | | | |
| R9 | CB AD/WR D6 | 30 | | | | |
| R10 | CB AD/WR D7 | 32 | 4 | 100 | pF | 1-4 |
| R11 | CB AD/WR D16 | 33 | 4 | 75 | pF | 1-4 |
| R12 | CB AD/WR D17 | 34 | | | | |
| R13 | CB AD/WR D18 | 35 | | | | |
| R14 | CB AD/WR D19 | 36 | 4 | 75 | pF | 1-4 |
| R15 | PWR UP MS CL | 44 | 2 | 75 | pF | 1-4 |
| R16 | GREFRESH REQ | 45 | 2 | 100 | pF | 1-4 |
| R17 | LPBACK SEL 1 | 71 | 10 | 200 | pF | 1-4 |
| R18 | LPBACK SEL 0 | 72 | 10 | 125 | pF | 1-4 |
| R19 | CB AD/WR D15 | 73 | 4 | 100 | pF | 1-4 |
| R20 | CB AD/WR D14 | 74 | 4 | 100 | pF | 1-4 |



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TABLE XVIII. Input and output capacitance test - Continued.

| Test cond | Signal name | Pin no. | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|
| R21 | CB AD/WR D13 | 76 | 4 | 100 | pF | 1-4 |
| R22 | CB AD/WR D12 | 77 | | | | |
| R23 | CB AD/WR D11 | 78 | | | | |
| R24 | CB AD/WR D10 | 79 | | | | |
| R25 | CB AD/WR D9 | 80 | | | | |
| R26 | CB AD/WR D8 | 82 | 4 | 100 | pF | 1-4 |
| R27 | COM BUS CLK | 83 | 2 | 75 | pF | 1-4 |
| R28 | CL SCRUB CTR | 84 | 2 | 75 | pF | 1-4 |
| R29 | TEST CLOCK | 85 | 4 | 75 | pF | 1-4 |
| R30 | ADVSCRUB CTR | 94 | 2 | 75 | pF | 1-4 |
| R31 | FIRST WORD | 96 | 6 | 125 | pF | 1-4 |
| R32 | MEM A/BUS A0 | 2 | 10 | 125 | pF | 1-4 |
| R33 | MEM A/BUS A1 | 3 | | | | |
| R34 | MEM A/BUS A2 | 4 | | | | |
| R35 | MEM A/BUS A3 | 5 | 10 | 125 | pF | 1-4 |
| R36 | MEM D/BUS 1 | 6 | 2 | 75 | pF | 1-4 |
| R37 | MEM D/BUS 3 | 7 | | | | |
| R38 | MEM D/BUS 5 | 8 | | | | |
| R39 | MEM D/BUS 7 | 9 | 2 | 75 | pF | 1-4 |
| R40 | MEM A/BUS A4 | 11 | 10 | 125 | pF | 1-4 |



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TABLE XVIII. Input and output capacitance test - Continued.

| Test cond | Signal name | Pin no. | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|
| R41 | MEM A/BUS A5 | 12 | 10 | 125 | pF | 1-4 |
| R42 | MEM A/BUS A6 | 13 | | | | |
| R43 | MEM A/BUS A7 | 14 | 10 | 125 | pF | 1-4 |
| R44 | MEM A/BUSA16 | 15 | 6 | 125 | pF | 1-4 |
| R45 | MEM A/BUSA17 | 16 | 6 | 125 | pF | 1-4 |
| R46 | MEM D/BUS 9 | 17 | 2 | 75 | pF | 1-4 |
| R47 | MEM D/BUS 11 | 18 | | | | |
| R48 | MEM D/BUS 13 | 19 | | | | |
| R49 | MEM D/BUS 15 | 20 | 2 | 75 | pF | 1-4 |
| R50 | MEM A/BUSB16 | 37 | 6 | 125 | pF | 1-4 |
| R51 | MEM A/BUSB17 | 38 | 6 | 150 | pF | 1-4 |
| R52 | MEM A/BUS B0 | 39 | 10 | 150 | pF | 1-4 |
| R53 | MEM A/BUS B1 | 41 | 10 | 125 | pF | 1-4 |
| R54 | MEM A/BUS B2 | 42 | 10 | 100 | pF | 1-4 |
| R55 | MEM A/BUS B3 | 43 | 10 | 125 | pF | 1-4 |
| R56 | MEM A/BUS B4 | 47 | | | | |
| R57 | MEM A/BUS B5 | 48 | | | | |
| R58 | MEM A/BUS B6 | 49 | | | | |
| R59 | MEM A/BUS B7 | 50 | 10 | 125 | pF | 1-4 |
| R60 | MEM A/BUS A8 | 52 | 8 | 100 | pF | 1-4 |



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TABLE XVIII. Input and output capacitance test - Continued.

| Test cond | Signal name | Pin no. | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|
| R61 | MEM A/BUS A9 | 53 | 8 | 100 | pF | 1-4 |
| R62 | MEM A/BUSA10 | 54 | | | | |
| R63 | MEM A/BUSA11 | 55 | 8 | 100 | pF | 1-4 |
| R64 | MEM D/BUS 6 | 56 | 2 | 75 | pF | 1-4 |
| R65 | MEM D/BUS 4 | 57 | | | | |
| R66 | MEM D/BUS 2 | 58 | | | | |
| R67 | MEM D/BUS 0 | 59 | 2 | 75 | pF | 1-4 |
| R68 | MEM A/BUSA12 | 61 | 8 | 125 | pF | 1-4 |
| R69 | MEM A/BUSA13 | 62 | 8 | 100 | pF | 1-4 |
| R70 | MEM A/BUSA14 | 63 | | | | |
| R71 | MEM A/BUSA15 | 64 | 8 | 100 | pF | 1-4 |
| R72 | MEM A/BUSA18 | 65 | 4 | 100 | pF | 1-4 |
| R73 | MEM A/BUSA19 | 66 | 4 | 150 | pF | 1-4 |
| R74 | MEM D/BUS 8 | 67 | 2 | 75 | pF | 1-4 |
| R75 | MEM D/BUS 10 | 68 | | | | |
| R76 | MEM D/BUS 12 | 69 | | | | |
| R77 | MEM D/BUS 14 | 70 | 2 | 75 | pF | 1-4 |
| R78 | MEM A/BUSB18 | 87 | 4 | 214 | pF | 1-4 |
| R79 | MEM A/BUSB19 | 88 | 4 | 75 | pF | 1-4 |
| R80 | MEM A/BUS B8 | 89 | 8 | 100 | pF | 1-4 |



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TABLE XVIII. Input and output capacitance test - Continued.

| Test cond | Signal name | Pin no. | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|
| R81 | MEM A/BUS B9 | 91 | 8 | 125 | pF | 1-4 |
| R82 | MEM A/BUSB10 | 92 | 8 | 100 | pF | 1-4 |
| R83 | MEM A/BUSB11 | 93 | | | | |
| R84 | MEM A/BUSB12 | 97 | | | | |
| R85 | MEM A/BUSB13 | 98 | 8 | 100 | pF | 1-4 |
| R86 | MEM A/BUSB14 | 99 | 8 | 125 | pF | 1-4 |
| R87 | MEM A/BUSB15 | 100 | 8 | 125 | pF | 1-4 |

NOTES:

1. Input and output capacitance test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Capacitance meter frequency | 1 MHz | ± 100 KHz | R1-R87 |
| Capacitance meter voltage | 15 mV | ± 1.5 mV | R1-R87 |
| Capacitance measurement | NA | ± 10.2 pF | R1 |
| | NA | ± 4.7 pF | R2 |
| | NA | ± 4.8 pF | R3-R10 |
| | NA | ± 3.55 pF | R11-R14 |
| | NA | ± 3.65 pF | R15 |
| | NA | ± 4.9 pF | R16 |



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NOTES: - Continued.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Capacitance measurement | NA | ± 9.5 pF | R17 |
| | NA | ± 5.75 pF | R18 |
| | NA | ± 4.8 pF | R19-R26 |
| | NA | ± 3.65 pF | R27,R28 |
| | NA | ± 3.55 pF | R29 |
| | NA | ± 3.65 pF | R30 |
| | NA | ± 5.95 pF | R31 |
| | NA | ± 5.75 pF | R32-R35 |
| | NA | ± 3.65 pF | R36-R39 |
| | NA | ± 5.75 pF | R40-R43 |
| | NA | ± 5.95 pF | R44,R45 |
| | NA | ± 3.65 pF | R46-R49 |
| | NA | ± 5.95 pF | R50 |
| | NA | ± 7.2 pF | R51 |
| | NA | ± 7.0 pF | R52 |
| | NA | ± 5.75 pF | R53 |
| | NA | ± 4.5 pF | R54 |
| | NA | ± 5.75 pF | R55-R59 |
| | NA | ± 4.6 pF | R60-R63 |
| | NA | ± 3.65 pF | R64-R67 |
| | NA | ± 5.85 pF | R68 |
| | NA | ± 4.6 pF | R69-R71 |
| | NA | ± 4.8 pF | R72 |
| | NA | ± 7.3 pF | R73 |
| | NA | ± 3.65 pF | R74-R77 |
| | NA | ± 10.5 pF | R78 |
| | NA | ± 3.55 pF | R79 |
| | NA | ± 4.6 pF | R80 |
| | NA | ± 5.85 pF | R81 |
| | NA | ± 4.6 pF | R82-R85 |
| | NA | ± 5.85 pF | R86,R87 |

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 12.

4. Abbreviations used are defined in 6.1.



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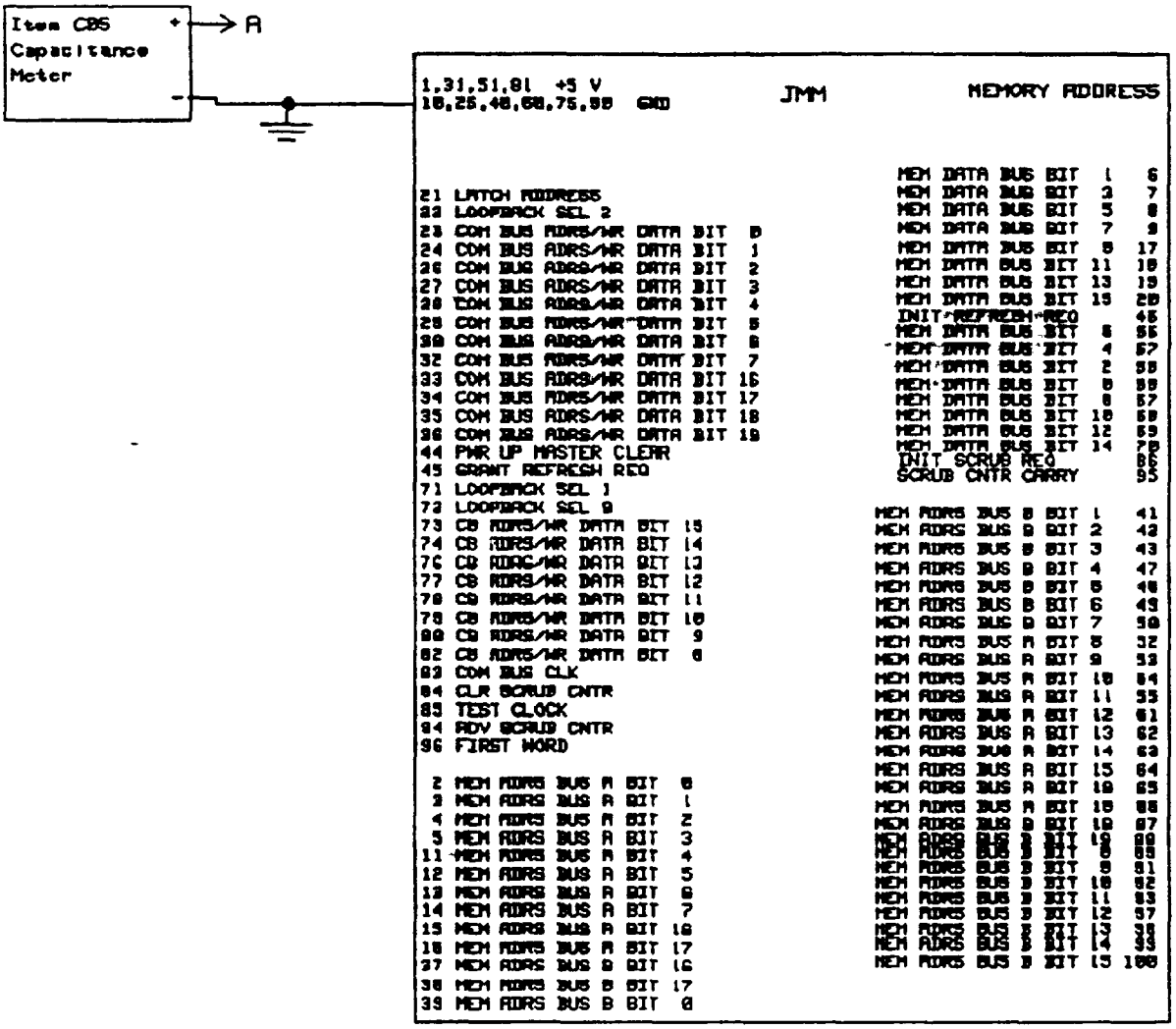



FIGURE 12. Input and output capacitance test circuit for table XVIII.



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TABLE XIX. Input clamp diode latchup test.

| Test cond | Signal name | Pin no. | Iin (mA) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| S | LATCH ADRES | 21 | -18.0 | -1.2 | 2.4 | V | 1-3 |

NOTES:

1. Input clamp diode latchup test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | Off | NA | S |
| Power supply current limit | NA | NA | S |
| Forcing current, (Iin) | -18 mA | ± 1.8 mA | S |
| Pattern voltage high, (VH)<br>low, (VL) | NA<br>NA | NA<br>NA | S<br>S |
| Input voltage measurement | NA | ± 60 mV | S |

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XIX and the voltage then measured.

(d) The forcing pulse shall have a duration of 500 ± 2 nS.

3. Abbreviations used are defined in 6.1.



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TABLE XX. Power supply transient amplitude test.

| Test cond | Parameter | Pin no. | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|
| T1 | PSTA | 1 | 0 | 400 | mV | 1-4 |
| T2 | PSTA | 51 | 0 | 400 | mV | 1-4 |

NOTES:

1. Power supply transient amplitude test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 5.0 V | ± 250 mV | T1,T2 |
| Power supply current limit | 2.45 A | NA | T1,T2 |
| Pattern voltage<br>high, (VH)<br>low, (VL) | <br>3.0 V<br>0.0 V | <br>± 100 mV<br>± 100 mV | <br>T1,T2<br>T1,T2 |
| Power supply transient amplitude measurement | NA | ± 20 mV | T1,T2 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.



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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 13 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 14.

4. Abbreviations used are defined in 6.1.



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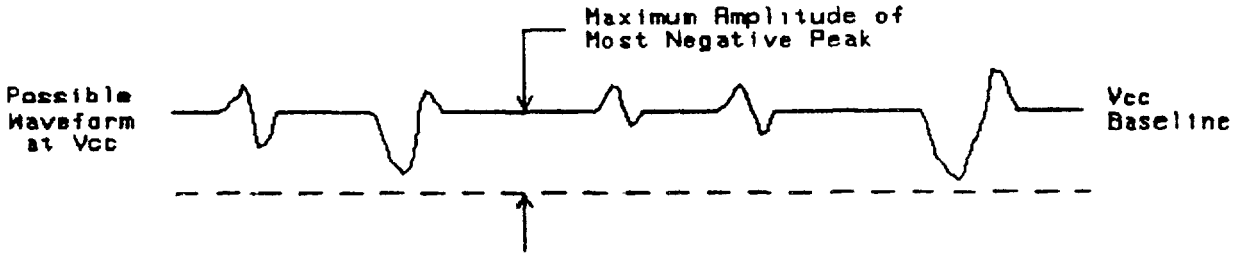
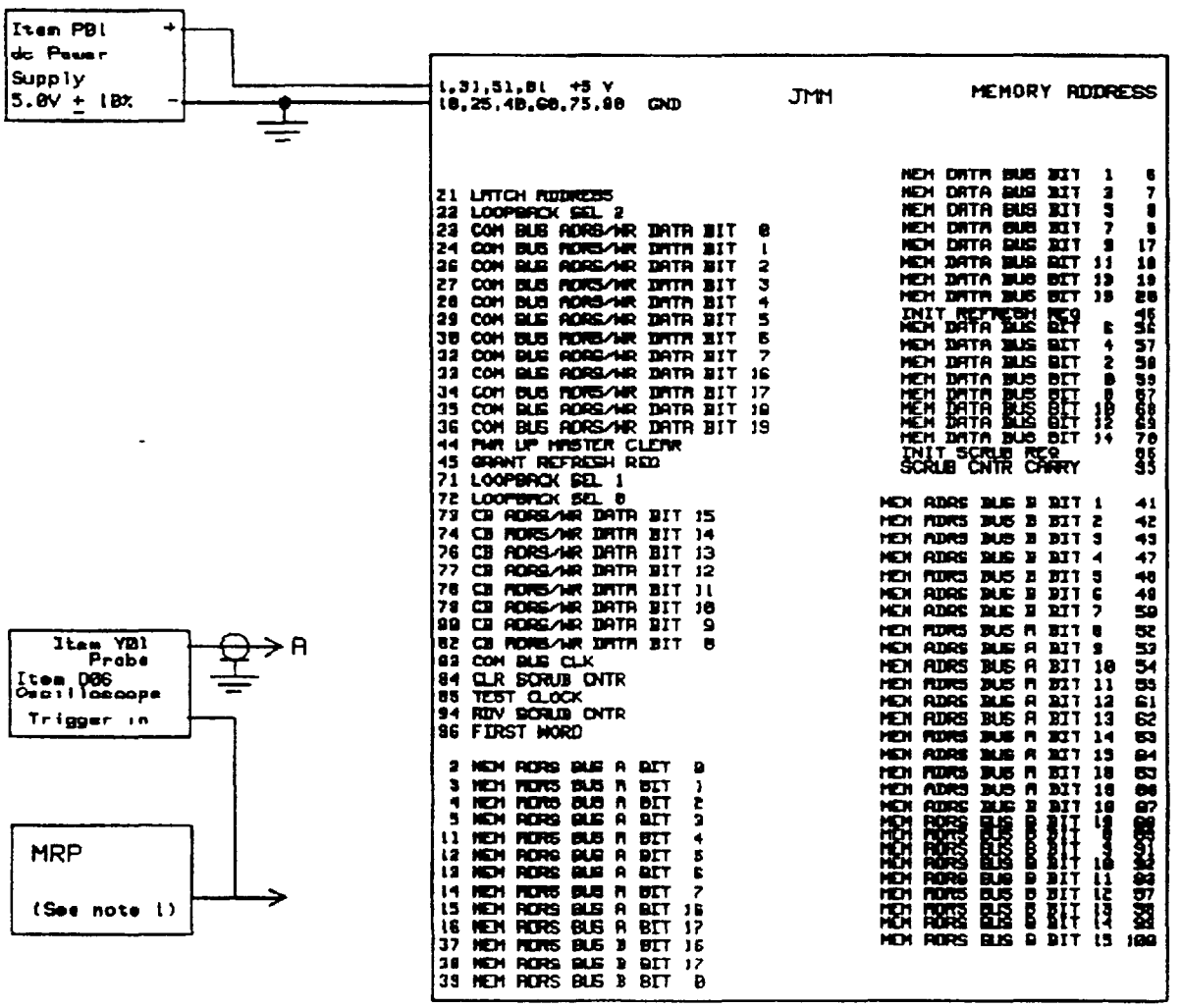



FIGURE 13. Power supply transient amplitude waveform for table XX.



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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 14. Power supply transient amplitude test circuit for table XX.



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